

# Shocks, Frictions, and Inequality in US Business Cycles<sup>†</sup>

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*We show how a heterogeneous agent New Keynesian (HANK) model with incomplete markets and portfolio choice can be estimated in state space using a Bayesian approach. To render estimation feasible, the structure of the economy can be exploited and the dimensionality of the model automatically reduced based on the Bayesian priors. We apply this approach to analyze how much inequality matters for the business cycle and vice versa. Even when the model is estimated on aggregate data alone and with a set of shocks and frictions designed to match aggregate data, it broadly reproduces observed US inequality dynamics. (JEL D31, D52, E12, E32, E52, E62)*

A new generation of monetary business cycle models with heterogeneous agents and incomplete markets (known as HANK models) has become popular. This new class of models implies new transmission channels of monetary policy<sup>1</sup> and fiscal policy<sup>2</sup> as well as new sources of business cycle fluctuations operating through household portfolio decisions.<sup>3</sup> Much of this literature to date has focused on the importance of household heterogeneity for specific transmission channels, shocks, or puzzles. Much emphasis has been placed on matching the stationary distribution of household characteristics, while the dynamic responses of the model at both the macro and micro levels have been left largely unrestricted by the data. This paper presents a novel approach that uses Bayesian techniques to estimate the dynamic properties of heterogeneous agent New Keynesian (HANK) models and fills this gap.

This is an important milestone because, first, it expands the data that can be used in the estimation of macroeconomic models to include the dynamics of cross-sectional

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<sup>1</sup>Auclert (2019) analyzes the redistributive effects of monetary policy; Kaplan, Moll, and Violante (2018) show the importance of indirect income effects; and Luetticke (2021) analyzes the portfolio rebalancing channel of monetary policy. McKay, Nakamura, and Steinsson (2016) examine the effectiveness of forward guidance.

<sup>2</sup>Auclert, Rognlie, and Straub (2018); Bayer, Born, and Luetticke (2023); and Hagedorn, Manovskii, and Mitman (2019) discuss fiscal multipliers. McKay and Reis (2016, 2021) discuss the role of automatic stabilizers.

<sup>3</sup>Bayer et al. (2019) quantify the importance of shocks to idiosyncratic income risk, and Guerrieri and Lorenzoni (2017) consider the effects of shocks to the borrowing limit.

observations at the household (and potentially firm) level. Additional data in the estimation of these models have far-reaching implications, from model selection to the identification of economic mechanisms. The data will also improve our understanding of the distributional consequences of the business cycle and the policy response to it. Second, our method allows these models to be rigorously tested and compared with the established representative agent literature. In this respect, our method provides an important bridge between the heterogeneous agent literature—with its focus on stationary distributions—and the business cycle literature—with its focus on aggregate shocks and frictions.

To remain in line with conventional methods for solving and estimating dynamic stochastic general equilibrium models, we propose to estimate HANK models in their state-space representation, too.<sup>4</sup> This leverages the experience that economists have with state-space methods and the vast tool set that has developed around them: for example, obtaining variance decompositions at business cycle frequencies (Uhlig 2001), using different sampling and filtering techniques (Acharya et al. 2021; Herbst and Schorfheide 2014, 2015), or dealing with mixed-frequency data. Our state-space approach also allows to translate established nonlinear solution methods (occasionally binding constraints (Guerrieri and Iacoviello 2015) or higher-order perturbations (Fernández-Villaverde et al. 2015)) easily to the HANK setting.

To obtain a state-space representation, we follow Reiter (2009) and linearize the model but then exploit two aspects for estimation. First, the economy can be written in a modular way, separating the components that only matter for the dynamics and the elements that reflect household heterogeneity. This implies that *relinearizing* the HANK model after a parameter change is as little numerically demanding as for the representative agent model (RANK) when estimating aggregate shocks and frictions. The modularity also allows us to provide a toolbox for estimating HANK models where the aggregate part can be easily customized. Second, we develop a novel model-reduction approach that emerges naturally from the Bayesian setting and drastically speeds up the *solution* of the linearized economy. We show that the speed and accuracy of the proposed solution method is comparable to the sequence-space approaches proposed by Auclert et al. (2021) and Boppart, Krusell, and Mitman (2018).

Concretely, we extend the method of Bayer and Luetticke (2020) with a more flexible and better-informed treatment of the value and distribution functions. This results in an arbitrarily precise approximation of the full model using information from the stationary equilibrium. Our novel Bayesian reduction technique then exploits the fact that we have prior information for model reduction, allowing us to go beyond what the literature has suggested. Specifically, we solve the model once under the parameterization implied by the mode of the prior distribution and use this solution to derive a factor representation of the heterogeneous agent part of the model. We show that this factor representation can be used for a very strong model reduction, even as the parameters change. It reduces the system of difference equations from initially more than 660,000 variables and equations to 2,900 in a first step based on the model's stationary equilibrium, and finally to fewer than 600 using the

<sup>4</sup> Auclert et al. (2021) propose an alternative technique for estimating heterogeneous agent models that solves the model in sequence space and uses the resulting MA- $\infty$  representation for estimation.

factor representation of the model's dynamics under the prior. The key here is the same insight that makes sequence-space methods feasible: household decisions are driven by a small set of prices and their limited and tractable dynamics.

As an application, we study a business cycle model in the spirit of Smets and Wouters (2007) and fuse it with the New Keynesian incomplete markets model of Bayer et al. (2019). This fused model then features capacity utilization, a frictional labor market with sticky wages and progressive taxation, as well as the battery of shocks that drive business cycle fluctuations in estimated New Keynesian models: aggregate and investment-specific productivity shocks, wage and price markup shocks, monetary and fiscal policy shocks, and risk premium shocks. To this battery of standard RANK model business cycle shocks, we add two additional incomplete market-specific shocks: shocks to the progressivity of taxes and shocks to idiosyncratic productivity risk. With this model at hand, we tackle three questions. First, to what extent does the inclusion of incomplete markets change our view of US business cycles? Second, can the model capture the dynamics of US inequality? Third, if so, which business cycle shocks and policies are important drivers?

In our model, precautionary motives play an important role for consumption-savings decisions. Since individual income is subject to idiosyncratic risk that cannot be directly insured and borrowing is constrained, households structure their savings decisions and portfolio allocations to optimally self-insure and achieve consumption smoothing. In particular, we assume that households can either hold liquid nominal bonds or invest in illiquid physical capital. Capital is illiquid since its market is segmented and households participate only from time to time. This portfolio-choice component, which gives rise to an endogenous liquidity premium, and the presence of occasional hand-to-mouth consumers lead the HANK model to have rich distributional dynamics in response to aggregate shocks.

To answer the first question of whether incomplete markets change our view of US business cycles, we estimate the HANK model using the same set of aggregate shocks and observables as in Smets and Wouters (2007), covering the period 1954 to 2019, and compare it to the representative household analog (RANK). We find that both models tell a similar story about the business cycle, but there are some differences because the incomplete market structure closely links aggregate consumption to the distributional consequences of shocks. In particular, technology shocks become more important for consumption growth in HANK, at the expense of markup shocks, because a significant fraction of households have few assets and a high marginal propensity to consume out of wage income.

These distributional consequences are also important for our second question. Here, we find that the HANK model can simultaneously account for the dynamics of the US business cycle and inequality between 1954 and 2019. Our model translates the business cycle shocks estimated from aggregate data into persistent movements in wealth and income inequality. These movements are broadly consistent with the U-shaped evolution of the wealth and income shares of the top 10 percent of US households in the data.<sup>5</sup>

<sup>5</sup>We focus on the top ten shares from the World Inequality Database because these measures are available since the 1950s and are most similar across alternative data sources, such as the Survey of Consumer Finances; see Kopczuk (2015) and Bricker et al. (2016).

Based on this finding, we then answer our third question, what drives US inequality, by reestimating the model. We now include the top ten shares as observables and allow for two additional, also observed, variables that directly affect the distribution of income: the progressivity of taxes and idiosyncratic income risk. We find that income risk partially replaces risk premium shocks in explaining aggregate consumption growth. Shocks to the progressivity of taxes have some importance for wealth inequality, as they persistently change the net income distribution and self-insurance incentives.

These findings are also reflected in historical decompositions of US inequality. We find that wealth inequality, as measured by the share held by the top 10 percent, is largely driven by two factors: shocks to investment technology and shocks to price markups. Shocks to technology have strong effects on asset prices and returns and through them have persistent effects on the distribution of wealth (as shown empirically in Kuhn, Schularick, and Steins 2020). Key to this are portfolio differences between the rich and the poor. The former hold their wealth in liquid, low-return assets, while the latter hold it in illiquid form. Price markup shocks work through the income distribution, and we estimate that the persistent rise in income inequality since the 1980s is related to higher price markups. However, there is some tension between the high volatility of markups in the model and the low cyclical volatility of the top income share.

To our knowledge, our paper is one of the first to provide an encompassing estimation of shocks and frictions using a HANK model with portfolio choice. Most of the literature on monetary heterogeneous agent models has used a calibration approach.<sup>6</sup> Auclert, Rognlie, and Straub (2020); Hagedorn, Manovskii, and Mitman (2018); and Bayer, Born, and Luetticke (2023) go beyond calibration but use a limited information approach based on impulse response function matching.

The inclusion of distributional data in the estimation is novel and potentially informative for identifying the sources of business cycle fluctuations. Our paper is related to Chang, Chen, and Schorfheide (2021) in the sense that it estimates a state-space model of both distributional (cross-sectional) data and aggregates. Chang, Chen, and Schorfheide (2021) find that shocks to the cross-sectional distribution of income, in the sense of an SVAR, have only a small impact on the aggregate time series. Our finding that structural estimates are relatively robust to the inclusion or exclusion of cross-sectional information is similar to their results.<sup>7</sup> The estimated muted importance of cross-sectional shocks is also consistent with the findings of Berger, Bocola, and DAVIS (2023), who use a business cycle accounting approach. Bilbiie, Primiceri, and Tambalotti (2022) estimate a tractable heterogeneous agent model in state-space form (two types of agents with stochastic transitions between types) using a full information approach and data on the cross-sectional dispersion

<sup>6</sup>See, for example, Auclert, Rognlie, and Straub (2018); Ahn et al. (2018); Bayer et al. (2019); Broer et al. (2019); Challe and Ragot (2015); Den Haan, Rendahl, and Riegler (2017); Ferriere and Navarro (forthcoming); Gornemann, Kuester, and Nakajima (2012); Guerrieri and Lorenzoni (2017); McKay, Nakamura, and Steinsson (2016); McKay and Reis (2016); Ravn and Sterk (2017); Sterk and Tenreyro (2018); and Wong (2019).

<sup>7</sup>Our approach is different from and simpler than the method proposed by Liu and Plagborg-Møller (2023), which includes full cross-sectional information in the estimation of a heterogeneous agent DSGE model. In contrast, we use the model only to fit certain generalized cross-sectional moments.

of labor earnings and income. They find amplification of aggregate shocks with heterogeneity.

Our findings add some insights to the literature on the drivers of inequality.<sup>8</sup> Kaymak and Poschke (2016) and Hubmer, Krusell, and Smith (2020) use quantitative models to study permanent changes in the US tax and transfer system and the variance of income. In terms of methods, these papers solve for steady-state transitions of calibrated models, while we estimate our model using US macro and micro time series data. They find that tax and transfer changes can explain a significant part of the recent increase in wealth inequality. Our paper is the first to quantify the distributional consequences of all standard business cycle shocks and to estimate their importance in explaining US inequality. In addition to business cycle shocks, our model incorporates changes in the US tax system and income risk and allows us to compare their relative importance for the evolution of income and wealth inequality. We find that business cycle shocks are important for wealth inequality through their effect on asset prices and returns.

The remainder of this paper is organized as follows. Section I describes our model economy, its sources of fluctuations and frictions. Section II provides details on the numerical solution method and the estimation technique. Section III presents our model variants, the parameters that we calibrate to match steady-state targets, prior and posterior distributions for the remaining parameters that we estimate, and an assessment of our solution approach based on the estimated posterior distribution. It also provides an overview of the data we use in our estimation. Section IV discusses the estimated shocks and frictions that drive the US business cycle and inequality dynamics. Section V concludes.

## I. Model

We model an economy composed of a firm sector, a household sector, and a government sector. Of these three sectors, only the household sector deviates from the standard New Keynesian DSGE model structure, as in Smets and Wouters (2007) or Christiano, Eichenbaum, and Evans (2005). In detail, the firm sector comprises (i) final goods producers who bundle the intermediate goods, (ii) intermediate goods producers who rent out labor services and capital from perfectly competitive markets but face monopolistic competition in the goods market as they produce differentiated goods and set prices, (iii) producers of capital goods who turn final goods into capital subject to adjustment costs, and (iv) labor packers who produce labor services combining differentiated labor from unions that differentiate raw labor rented out from households. Price setting for the intermediate goods as well as wage setting by unions is subject to a friction *a la* Calvo (1983).

Households earn income from supplying (raw) labor and capital and from owning the firm sector, absorbing all its rents that stem from the market power of unions and intermediate goods producers, and decreasing returns to scale in capital goods production. They face idiosyncratic income risk against which they can self-insure

<sup>8</sup>There is a growing literature on inequality dynamics: on the theoretical side, e.g., Gabaix et al. (2016) and on the empirical side, e.g., Heathcote, Perri, and Violante (2010); Piketty and Saez (2003); or Saez and Zucman (2016).

by trading liquid and illiquid assets, giving rise to endogenous heterogeneity. We also consider a representative agent variant with full insurance.

The government sector runs both a fiscal authority and a monetary authority. The fiscal authority levies progressive taxes on labor incomes and profits, issues government bonds, and adjusts expenditures to stabilize debt in the long run and aggregate demand in the short run. The monetary authority sets the nominal interest rate on government bonds according to a Taylor rule.

### A. Firm Sector

Since the firm sector involves dynamic decisions, we need to make an assumption about the discount factor used in these decisions. With household heterogeneity, stochastic discount factors across households might differ. For this reason, we make the simplifying assumption that the firm sector is run by managers who are risk neutral, have no asset market access, but have the same time preferences as households.<sup>9</sup> Managers are a mass-zero group in the economy so that their consumption does not show up in any resource constraint and as a result, all profits of the firm sector go to households.

*Final Goods Producers.*—Final goods producers bundle varieties  $j$  of differentiated goods according to the Dixit-Stiglitz aggregator

$$(1) \quad Y_t = \left( \int y_{jt}^{\frac{\eta_t-1}{\eta_t}} dj \right)^{\frac{\eta_t}{\eta_t-1}},$$

with elasticity of substitution  $\eta_t$ . Each of these differentiated goods is offered at price  $p_{jt}$ , so that the aggregate price level is given by  $P_t = \left( \int p_{jt}^{1-\eta_t} dj \right)^{1/(1-\eta_t)}$  and the demand for each of the varieties is

$$(2) \quad y_{jt} = \left( \frac{p_{jt}}{P_t} \right)^{-\eta_t} Y_t.$$

*Intermediate Goods Producers.*—Intermediate goods are produced with a constant returns to scale production function

$$(3) \quad Y_{jt} = Z_t N_{jt}^\alpha (u_{jt} K_{jt})^{(1-\alpha)},$$

where  $\alpha$  is the labor share in production,  $Z_t$  is total factor productivity that follows an autoregressive process in logs,  $N_{jt}$  is the labor bundle firm  $j$  hires at time  $t$ , and  $u_{jt} K_{jt}$  are capital services taking into account utilization  $u_{jt}$ , i.e., the intensity with which the capital stock  $K_{jt}$  is used. An intensity higher than normal results in increased depreciation of capital according to  $\delta(u_{jt}) = \delta_0 + \delta_1(u_{jt} - 1) + \delta_2/2(u_{jt} - 1)^2$ , which, assuming  $\delta_1, \delta_2 > 0$ , is an increasing and convex function of utilization. Without loss of generality, capital utilization in steady state is normalized to 1, so that  $\delta_0$  denotes the steady-state depreciation rate of capital goods.

<sup>9</sup> Since we solve the model by a first-order perturbation in aggregate shocks, fluctuations in stochastic discount factors are irrelevant.



Given demand, the producer minimizes costs,  $w_t^F N_t - [r_t + q_t \delta(u_{jt})] K_t$ , where  $r_t$  and  $q_t$  are the rental rate and the (producer) price of capital goods, respectively, and  $w_t^F$  is the real wage the firm faces. Factor markets are perfectly competitive. Hence, the first-order conditions for labor and effective capital read

$$(4) \quad w_t^F = \alpha mc_{jt} Z_t \left( \frac{u_{jt} K_{jt}}{N_{jt}} \right)^{1-\alpha}$$

and

$$(5) \quad r_t + q_t \delta(u_{jt}) = u_{jt} (1 - \alpha) mc_{jt} Z_t \left( \frac{N_{jt}}{u_{jt} K_{jt}} \right)^\alpha,$$

where  $mc_{jt}$  is the marginal cost of firm  $j$ . The optimal utilization is given by

$$(6) \quad q_t [\delta_1 + \delta_2(u_{jt} - 1)] = (1 - \alpha) mc_{jt} Z_t \left( \frac{N_{jt}}{u_{jt} K_{jt}} \right)^\alpha.$$

Combining the three first-order conditions, it is easy to see that given the constant returns to scale production function, marginal costs are constant across producers  $mc_{jt} = mc_t$ .

We assume that intermediate goods producers face price adjustment frictions a la Calvo (1983); the firms' managers maximize the present value of real profits subject to this price adjustment friction and the demand curve (2). They hence maximize

$$(7) \quad E_0 \sum_{t=0}^{\infty} \beta^t \lambda_Y^t (1 - \tau_t^L) Y_t^{1-\tau_t^P} \left[ \left( \frac{p_{jt} \bar{\pi}_Y^t}{P_t} - mc_t \right) \left( \frac{p_{jt} \bar{\pi}_Y^t}{P_t} \right)^{-\eta_t} \right]^{1-\tau_t^P},$$

with a time-constant discount factor  $\beta$ . Prices are indexed to the steady-state inflation rate  $\bar{\pi}$  and can be discretionally adjusted price with probability  $1 - \lambda_Y$ . The parameters  $\tau_t^P$  and  $\tau_t^L$  characterize the progressivity and level of the tax schedule, which we discuss in more detail when describing the household sector.

The corresponding first-order condition for price setting implies a Phillips curve,

$$(8) \quad \log\left(\frac{\pi_t}{\bar{\pi}}\right) = \beta E_t \log\left(\frac{\pi_{t+1}}{\bar{\pi}}\right) + \kappa_Y \left( mc_t - \frac{1}{\mu_t^Y} \right),$$

where we dropped all terms irrelevant for a first-order approximation and defined  $\kappa_Y = [(1 - \lambda_Y)(1 - \lambda_Y \beta)] / \lambda_Y$ . Here,  $\pi_t$  is the gross inflation rate of final goods,  $\pi_t := P_t / P_{t-1}$ ,  $mc_t := MC_t / P_t$  is the real marginal costs, and  $\mu_t^Y = \eta_t / (\eta_t - 1)$  is the target markup. This target fluctuates in response to markup shocks,  $\epsilon_t^{\mu Y}$ , and follows a log AR(1) process.

*Capital Goods Producers.*—Capital goods producers take the relative price of capital goods,  $q_t$ , as given in deciding about their output; i.e., they maximize

$$(9) \quad E_0 \sum_{t=0}^{\infty} \beta^t I_t \left\{ \Psi_t q_t \left[ 1 - \frac{\phi}{2} \left( \log \frac{I_t}{I_{t-1}} \right)^2 \right] - 1 \right\},$$

where  $\phi$  controls the strength of the quadratic investment adjustment costs and  $\Psi_t$  governs the marginal efficiency of investment a la Justiniano, Primiceri, and Tambalotti (2011), which follows an AR(1) process in logs and is subject to shocks  $\epsilon_t^{\Psi}$ .<sup>10</sup>

Optimality of the capital goods production requires (again dropping all terms irrelevant up to first order)

$$(10) \quad \Psi_t q_t \left( 1 - \phi \log \frac{I_t}{I_{t-1}} \right) = 1 - \beta E_t \left[ \Psi_{t+1} q_{t+1} \phi \log \left( \frac{I_{t+1}}{I_t} \right) \right],$$

and each capital goods producer will adjust its production until (10) is fulfilled.

Since all capital goods producers are symmetric, we obtain as the law of motion for aggregate capital

$$(11) \quad K_t - [1 - \delta(u_t)] K_{t-1} = \Psi_t \left[ 1 - \frac{\phi}{2} \left( \log \frac{I_t}{I_{t-1}} \right)^2 \right] I_t.$$

The functional form assumption implies that investment adjustment costs are minimized and equal to zero in the steady state.

*Labor Packers and Unions.*—Workers sell their labor services to a mass-one continuum of unions indexed by  $j$ , each of which offers a different variety of labor to labor packers, who then provide labor services to intermediate goods producers. Labor packers produce final labor services according to the production function

$$(12) \quad N_t = \left( \int \hat{n}_{jt}^{\frac{\zeta_t-1}{\zeta_t}} dj \right)^{\frac{\zeta_t}{\zeta_t-1}}$$

out of labor varieties  $\hat{n}_{jt}$  with elasticity of substitution  $\zeta_t$ . Cost minimization by labor packers implies that each variety of labor, each union  $j$ , faces a downward-sloping demand curve

$$(13) \quad \hat{n}_{jt} = \left( \frac{W_{jt}}{W_t^F} \right)^{-\zeta_t} N_t,$$

where  $W_{jt}$  is the *nominal* wage set by union  $j$  and  $W_t^F$  is the *nominal* wage at which labor packers sell labor services to intermediate goods producers.

Since unions have market power, they pay the households a wage lower than the price at which they sell labor to labor packers. Given the nominal wage  $W_t$  at which they buy labor from households and given the nominal wage index  $W_t^F$ , unions seek to maximize their discounted stream of profits. However, they face a Calvo (1983)-type adjustment friction with indexation, where  $\lambda_w$  is the probability to keep wages constant. They therefore maximize

$$(14) \quad E_0 \sum_{t=0}^{\infty} \beta^t \lambda_w^t \frac{W_t^F}{P_t} N_t \left[ \left( \frac{W_{jt} \bar{\pi}_W^t}{W_t^F} - \frac{W_t}{W_t^F} \right) \left( \frac{W_{jt} \bar{\pi}_W^t}{W_t^F} \right)^{-\zeta_t} \right]$$

<sup>10</sup>This shock has to be distinguished from a shock to the relative price of investment, which has been shown in the literature (Justiniano, Primiceri, and Tambalotti 2011; Schmitt-Grohé and Uribe 2012) to not be an important driver of business cycles as soon as one includes the relative price of investment as an observable. We therefore focus on the marginal efficiency of investment shock.



by setting  $W_{jt}$  in period  $t$  and keeping it constant except for indexation to  $\bar{\pi}_w$ , the steady-state wage inflation rate.

Since all unions are symmetric, we focus on a symmetric equilibrium and obtain the linearized wage Phillips curve from the corresponding first-order condition as follows, leaving out all terms irrelevant at a first-order approximation around the stationary equilibrium

$$(15) \quad \log\left(\frac{\pi_t^W}{\bar{\pi}_w}\right) = \beta E_t \log\left(\frac{\pi_{t+1}^W}{\bar{\pi}_w}\right) + \kappa_w \left( mc_t^w - \frac{1}{\mu_t^w} \right),$$

with  $\pi_t^W := W_t^F/W_{t-1}^F = (w_t^F/w_{t-1}^F)\pi_t^Y$  being wage inflation;  $w_t$  and  $w_t^F$  being the respective *real* wages for households and firms;  $mc_t^w = w_t/w_t^F$  being the actual and  $1/\mu_t^w = (\zeta_t - 1)/\zeta_t$  being the target markdown of wages the unions pay to households,  $W_t$ , relative to the wages charged to firms,  $W_t^F$ ; and  $\kappa_w = [(1 - \lambda_w)(1 - \lambda_w\beta)]/\lambda_w$ . This target fluctuates in response to markup shocks,  $\epsilon_t^{\mu W}$ , and follows a log AR(1) process.

### B. Households

There is a continuum of ex ante identical households of measure one, indexed by  $i$ . Households are infinitely lived, have time-separable preferences with time-discount factor  $\beta$ , and derive felicity from consumption  $c_{it}$  and leisure. They obtain income from supplying labor,  $n_{it}$ , from renting out capital,  $k_{it}$ , and from earning interest on bonds,  $b_{it}$ . What is more, they receive profits of firms,  $\Pi_t^Y = (1 - mc_t)Y_t$ , and unions,  $\Pi_t^U = (w_t^F - w_t)N_t$ . Households pay taxes on labor and profit income. Our baseline model features household heterogeneity. Households differ in their productivity and in whether they obtain profit income. They face incomplete markets in this baseline, and capital as an asset is illiquid, while bonds are liquid. For comparison, we also consider a representative agent variant.

*Preferences.*—With respect to leisure and consumption, households have Greenwood, Hercowitz, and Huffman (1988)—hereafter, GHH—preferences and maximize the discounted sum of felicity,

$$(16) \quad E_0 \max_{\{c_{it}, n_{it}\}} \sum_{t=0}^{\infty} \beta^t u[c_{it} - G(h_{it}, n_{it})].$$

The maximization is subject to the budget constraints described further below. The felicity function  $u$  exhibits a constant relative risk aversion with risk aversion parameter  $\xi$ ,

$$(17) \quad u(x_{it}) = \frac{x_{it}^{1-\xi} - 1}{1 - \xi},$$

where  $x_{it} = c_{it} - G(h_{it}, n_{it})$  is household  $i$ 's composite demand for goods consumption  $c_{it}$  and leisure and  $G$  measures the disutility from work. While  $n_{it}$  denotes a household's labor supply,  $h_{it}$  is the household's labor productivity.<sup>11</sup>

<sup>11</sup>The assumption of GHH preferences is mainly motivated by the fact that many estimated DSGE models of business cycles find small aggregate wealth effects in the labor supply; see, e.g., Schmitt-Grohé and Uribe (2012)

Assuming a (progressive) income tax schedule (which we borrow from Benabou 2002; Heathcote, Storesletten, and Violante 2017), a household's net labor income,  $y_{it}$ , is given by

$$(18) \quad y_{it} = (1 - \tau_t^L)(w_t h_{it} n_{it})^{1-\tau_t^P},$$

where  $w_t$  is the aggregate real wage rate and  $\tau_t^L$  and  $\tau_t^P$  determine the level and progressivity of the tax code. Given net labor income, the first-order condition for labor supply is

$$(19) \quad \frac{\partial G(h_{it}, n_{it})}{\partial n_{it}} = (1 - \tau_t^P)(1 - \tau_t^L)(w_t h_{it})^{1-\tau_t^P} n_{it}^{-\tau_t^P} = (1 - \tau_t^P) \frac{y_{it}}{n_{it}}.$$

Assuming that  $G$  has a constant elasticity with respect to  $n$ ,  $\partial G(h_{it}, n_{it})/\partial n_{it} = (1 + \gamma)[G(h_{it}, n_{it})/n_{it}]$  with  $\gamma > 0$ , we can simplify the expression for the composite consumption good,  $x_{it}$ , making use of this first-order condition (19), and substitute  $G(h_{it}, n_{it})$  out of the individual planning problem,

$$(20) \quad x_{it} = c_{it} - G(h_{it}, n_{it}) = c_{it} - \frac{1 - \tau_t^P}{1 + \gamma} y_{it}.$$

When the Frisch elasticity of labor supply is constant and the tax schedule has the form (18), the disutility of labor is always a fraction of labor income and constant across households. Therefore, in both the household's budget constraint and felicity function, only after-tax income enters, and neither hours worked nor productivity appear separately.

Aggregate effective labor supply depends on the distribution of  $h_{it}$  and tax progressivity. Without loss of generality, we assume  $G(h_{it}, n_{it}) = h_{it}^{1-\bar{\tau}^P} [n_{it}^{1+\gamma}/(1 + \gamma)]$ , where  $\bar{\tau}^P$  is the stationary equilibrium level of progressivity of the tax code. This functional form simplifies the household problem, as  $h_{it}$  drops out from the first-order condition as long as tax progressivity is constant. Then, all households supply the same number of hours  $n_{it} = N(w_t)$ . Total effective labor input,  $\int n_{it} h_{it} di$ , is then hence also equal to  $N(w_t)$  because we normalize  $\int h_{it} di = 1$ . Individual after-tax labor income is then

$$(21) \quad y_{it} = (1 - \tau_t^L)(w_t h_{it} n_{it})^{1-\bar{\tau}^P} = (1 - \tau_t^L)^{\frac{1+\gamma}{\gamma+\bar{\tau}^P}} (1 - \bar{\tau}^P)^{\frac{1-\bar{\tau}^P}{\gamma+\bar{\tau}^P}} w_t^{\frac{1+\gamma}{\gamma+\bar{\tau}^P} (1-\bar{\tau}^P)} h_{it}^{1-\bar{\tau}^P}.$$

In an extension, we allow for tax progressivity to vary over time because it can directly affect the distribution of net incomes and thereby wealth.<sup>12</sup>

*Baseline: Heterogeneous Households and Incomplete Markets.*—In our baseline, the household sector is subdivided into two types of agents: workers and

and Born and Pfeifer (2014). In online Appendix B.1, we show how the assumption of King, Plosser, and Rebelo (1988)—hereafter, KPR—preferences affects results. We also briefly discuss them in Section IVC.

<sup>12</sup>In this case, the parameter  $\tau_t^P$  governing the progressivity of the tax schedule evolves according to

$$(33a) \quad \frac{\tau_t^P}{\bar{\tau}^P} = \left( \frac{\tau_{t-1}^P}{\bar{\tau}^P} \right)^{\rho_P} \epsilon_t^P,$$

entrepreneurs. The transition between both types is stochastic. On top, workers face idiosyncratic labor productivity risk. Both, workers and entrepreneurs, rent out physical capital, but only workers supply labor. Entrepreneurs do not work but earn all pure rents in our economy except for the rents of unions, which are equally distributed across workers. All households self-insure against the income risks they face by saving in a liquid nominal asset (bonds) and a less liquid asset (capital). Trading these illiquid assets is subject to random participation in the capital market.

We assume that productivity evolves according to a log AR(1) process and a fixed probability of transition between the worker and the entrepreneur state:

$$(23) \quad \tilde{h}_{it} = \begin{cases} \exp(\rho_h \log \tilde{h}_{it-1} + \epsilon_{it}^h), & \text{with probability } 1 - \zeta \text{ if } h_{it-1} \neq 0; \\ 1, & \text{with probability } \iota \text{ if } h_{it-1} = 0; \\ 0, & \text{otherwise,} \end{cases}$$

with individual productivity  $h_{it} = \tilde{h}_{it} / \int \tilde{h}_{it} di$  such that  $\tilde{h}_{it}$  is scaled by its cross-sectional average,  $\int \tilde{h}_{it} di$ , to make sure that average worker productivity is constant. The shocks  $\epsilon_{it}^h$  to productivity are normally distributed with variance  $\bar{\sigma}_h^2$ .<sup>13</sup> With probability  $\zeta$  households become entrepreneurs ( $h = 0$ ). With probability  $\iota$  an entrepreneur returns to the labor force with median productivity. Besides their labor income, workers receive a share in union rents,  $\Pi_t^U$ , which are distributed lump-sum, leading to labor-income compression. For tractability, we assume union profits to be taxed at a fixed rate independent of the recipient's labor income.

For the distribution of firm profits (aside from union profits), we assume that they primarily go to entrepreneurs. However, entrepreneurs as a group can sell claims to a fraction  $\omega^\Pi$  of their profits as shares. These claims have stochastic maturity and are liquid. This stochastic maturity ensures finite prices for profit claims even at zero

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where  $\epsilon_t^P$  are shocks to tax progressivity. When tax progressivity does not coincide with its stationary equilibrium value, individual hours worked differ across agents and are given by

$$(19a) \quad n_{it} = \left[ (1 - \tau_t^P)(1 - \tau_t^L) \right]^{\frac{1}{\gamma + \tau_t^P}} \frac{\tau_t^P - \tau_t^L}{h_{it}^{\gamma + \tau_t^P}} \frac{1 - \tau_t^P}{w_t^{\gamma + \tau_t^P}},$$

such that aggregate effective hours are given by

$$(19b) \quad N_t = \int n_{it} h_{it} = \left[ (1 - \tau_t^P)(1 - \tau_t^L) \right]^{\frac{1}{\gamma + \tau_t^P}} \frac{1 - \tau_t^P}{w_t^{\gamma + \tau_t^P}} \underbrace{\int h_{it}^{\frac{\gamma + \tau_t^P}{\gamma + \tau_t^P}}}_{=: H_t}.$$

Here,  $H_t$  measures how the tax progressivity influences the (hours-weighted) average labor productivity. Scaling of the disutility of labor by  $h_{it}^{1 - \frac{\gamma + \tau_t^P}{\gamma + \tau_t^L}}$  is thus a normalization of  $H_t$  to one in the stationary equilibrium. Household after-tax labor income, plugging in the optimal supply of hours, is then

$$(22) \quad y_{it} = (1 - \tau_t^L)(w_t h_{it} n_{it})^{1 - \tau_t^P} = (1 - \tau_t^L)^{\frac{1 + \gamma}{\gamma + \tau_t^L}} (1 - \tau_t^P)^{\frac{1 - \tau_t^P}{\gamma + \tau_t^L}} \frac{1 + \gamma}{w_t^{\gamma + \tau_t^L}} (1 - \tau_t^P)^{\frac{\gamma + \tau_t^P}{\gamma + \tau_t^L}} h_{it}^{\frac{\gamma + \tau_t^P}{\gamma + \tau_t^L} (1 - \tau_t^P)}.$$

<sup>13</sup>In our baseline, we treat the variance as time fixed. We consider an extension where income risk follows a log AR(1) process with endogenous feedback to aggregate output growth:

$$(23a) \quad \sigma_{h,t}^2 = \bar{\sigma}_h^2 \exp(\hat{s}_t),$$

$$\hat{s}_{t+1} = \rho_s \hat{s}_t + \Sigma_Y \frac{Y_{t+1}}{Y_t} + \epsilon_t^\sigma;$$

i.e., at time  $t$  households observe a change in the variance of shocks that drive the next period's productivity.

interest rates of liquid assets. Each period  $\iota^\Pi$  claims mature. When a claim matures, it loses value and is replaced by a new issuance by the entrepreneurs. We assume a unit mass of profit shares, which then trade at price  $q_t^\Pi$ . The entrepreneurs then receive in each period the sum of the profits they have not sold plus the value of the new shares they sell:  $\Pi_t^E = (1 - \omega^\Pi)\Pi_t^F + \iota^\Pi q_t^\Pi$ .<sup>14</sup>

This modeling strategy allows us to match the income and wealth distribution following the idea by Castaneda, Diaz-Gimenez, and Rios-Rull (1998), while limiting the impact of profits on investment behavior and asset markets.

Given incomes, households optimize intertemporally subject to their budget constraint,

$$(24) \quad c_{it} + b_{it+1} + q_t k_{it+1} = b_{it} \frac{R_{it}}{\pi_t} + (q_t + r_t)k_{it} + y_{it} + \mathbf{1}\{h_{it} \neq 0\}(1 - \tau_t)\Pi_t^U \\ + \mathbf{1}\{h_{it} = 0\}(1 - \tau_t^L)(\Pi_t^E)^{1-\bar{\tau}^P},$$

$$k_{it+1} \geq 0, \quad b_{it+1} \geq \underline{B},$$

where  $\Pi_t^U$  is union profits taxed at the average tax rate  $\tau_t$ ;  $\Pi_t^E$  is profit payouts to entrepreneurs;  $b_{it}$  is real liquid assets;  $k_{it}$  is the amount of illiquid assets;  $q_t$  is the price of these assets;  $r_t$  is their dividend;  $\pi_t = P_t/P_{t-1}$  is realized inflation; and  $R_{it}$  is the nominal interest rate on liquid assets, which depends on whether the household is a borrower or lender, the efficiency of intermediation, the returns on profit shares, and the central bank's interest rate  $R_t^b$ , which is set one period before. All households that do not participate in the capital market ( $k_{it+1} = k_{it}$ ) still obtain dividends and can adjust their liquid asset holdings. Depreciated capital has to be replaced for maintenance, such that the dividend,  $r_t$ , is the net return on capital. Holdings of bonds have to be above an exogenous debt limit  $\underline{B}$ , and holdings of capital have to be nonnegative.

Substituting the expression  $c_{it} = x_{it} + [(1 - \bar{\tau}^P)/(1 + \gamma)]y_{it}$  for consumption, we obtain the budget constraint for the composite leisure-consumption good,

$$(25) \quad x_{it} + b_{it+1} + q_t k_{it+1} = b_{it} \frac{R_{it}}{\pi_t} + (q_t + r_t)k_{it} + \frac{\bar{\tau}^P + \gamma}{1 + \gamma} y_{it} \\ + \mathbf{1}\{h_{it} \neq 0\}(1 - \tau_t)\Pi_t^U \\ + \mathbf{1}\{h_{it} = 0\}(1 - \tau_t^L)(\Pi_t^E)^{1-\bar{\tau}^P},$$

$$k_{it+1} \geq 0, b_{it+1} \geq \underline{B}.$$

Households make their savings and portfolio choice between liquid bonds and illiquid capital in light of a capital market friction that renders capital illiquid because participation in the capital market is random and i.i.d. in the sense that only a fraction,  $\lambda$ , of households are selected to be able to adjust their capital holdings in a given period.

<sup>14</sup>Boar and Midrigan (2019) use a similar structure, where entrepreneurs retain a fraction of firm profits and thus, the size of markups has an impact on inequality in the economy.

Ex post returns  $R_{it}$  on the liquid asset are given by the average return of the liquid asset portfolio, composed of government bonds  $B_t$  and profit shares with a value of  $q_t^\Pi$ , i.e.,

$$(26) \quad R_{it} = \begin{cases} A_t \frac{R_t^b B_t + \pi_t [(1 - \iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F]}{B_t + q_{t-1}^\Pi}, & \text{if } b_{it} \geq 0; \\ A_t \frac{R_t^b B_t + \pi_t [(1 - \iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F]}{B_t + q_{t-1}^\Pi} + \bar{R}, & \text{if } b_{it} < 0. \end{cases}$$

The shifter  $A_t$  reflects a “risk-premium shock” (as in, for example, Smets and Wouters 2007) and is technically modeled as an intermediation efficiency here. The first part of the sum in the numerator is the interest payments on government bonds issued and bought in the previous period; the second part is the returns from selling the nonmatured profit claims and the share of profits that is paid out to shareholders. The denominator is the sum of the value of bonds and profit shares bought in the previous period.

Since a household’s saving decision— $(b'_a, k')$  for the case of adjustment and  $(b'_n, k)$  for nonadjustment—will be some nonlinear function of that household’s wealth and productivity, inflation and all other prices will be functions of the joint distribution,  $\Theta_t$ , of  $(b, k, h)$  in  $t$ . This makes  $\Theta$  a state variable of the household’s planning problem, and this distribution evolves as a result of the economy’s reaction to aggregate shocks. For simplicity, we summarize all effects of aggregate state variables, including the distribution of wealth and income, by writing the dynamic planning problem with time-dependent continuation values.

This leaves us with three functions that characterize the household’s problem: value function  $V^a$  for the case where the household adjusts its capital holdings, the function  $V^n$  for the case in which it does not adjust, and the expected continuation value,  $\mathbb{W}$ , over both,

$$(27) \quad \begin{aligned} V_t^a(b, k, h) &= \max_{b'_a, k'} u[x(b, b'_a, k, k', h)] + \beta E_t \mathbb{W}_{t+1}(b'_a, k', h'), \\ V_t^n(b, k, h) &= \max_{b'_n} u[x(b, b'_n, k, k, h)] + \beta E_t \mathbb{W}_{t+1}(b'_n, k, h'), \\ \mathbb{W}_{t+1}(b', k', h') &= \lambda V_{t+1}^a(b', k', h') + (1 - \lambda) V_{t+1}^n(b', k', h'). \end{aligned}$$

Expectations about the continuation value are taken with respect to all stochastic processes conditional on the current states. Maximization is subject to the corresponding budget constraint. The distribution  $\Theta_t$  then evolves according to

$$(28) \quad \begin{aligned} \Theta_{t+1}(b', k', h') &= \lambda \int_{b'=b_{a,t}^*(b, k, h), k'=k_t^*(b, k, h)} \Phi(h, h') d\Theta_t(b, k, h) \\ &\quad + (1 - \lambda) \int_{b'=b_{n,t}^*(b, k, h), k'=k} \Phi(h, h') d\Theta_t(b, k, h), \end{aligned}$$

where  $\Phi(\cdot)$  is the transition probability for  $h$ , and  $b_{n,t}^*$ ,  $b_{a,t}^*$ , and  $k_t^*$  are the time- $t$  optimal policies.

Importantly, following Reiter (2009), one can view the discretized version of (27) and (28) as a set of equations that pins down the dynamics of the value functions

and optimal policy for each  $b \times k \times h$  node as well as the transition of the mass of households at each of the nodes.

*Model Variant with a Representative Household and Complete Markets.*—With complete markets, when all households are homogeneous with equal and constant labor productivity  $h_i = 1$  and equally obtain all profit incomes, the planning problem is described by the consumption Euler equation for bonds instead of the above-mentioned set of equations. For an optimal consumption-savings policy,

$$(29) \quad u_c(x_t) = \beta E_t \frac{A_t R_t^b}{\pi_{t+1}} u_c(x_{t+1})$$

needs to hold, replacing (27). Again,  $x_t = c_t - G(n_t)$  is the composite consumption-leisure good.

The law of motion for the distribution (28) is replaced by the wealth accumulation equation given by the budget constraint

$$(30) \quad q_t K_{t+1} + B_{t+1} = \frac{R_t^b}{\pi_t} B_t + (q_t + r_t) K_t + (1 - \tau_t) \left[ \frac{\tau_t^p + \gamma}{1 + \gamma} (w_t N_t)^{(1 - \tau_t^p)} + \Pi_t^U + \Pi_t^F \right] - x_t,$$

and the consumption Euler equation for capital

$$(31) \quad u_c(x_t) = \beta E_t \frac{q_{t+1} + r_{t+1}}{q_t} u_c(x_{t+1}),$$

which then yields the optimal portfolio combination of  $K$  and  $B$  given return expectations.

### C. Government

This leaves us with the government sector. The government operates a monetary and a fiscal authority. The monetary authority controls the nominal interest rate on liquid assets, while the fiscal authority issues government bonds to finance deficits, chooses both the average tax rate in the economy and the tax progressivity, and makes expenditures.

We assume that monetary policy sets the nominal interest rate following a Taylor (1993)-type rule with interest rate smoothing:

$$(32) \quad \frac{R_{t+1}^b}{\bar{R}^b} = \left( \frac{R_t^b}{\bar{R}^b} \right)^{\rho_R} \left( \frac{\pi_t}{\bar{\pi}} \right)^{(1 - \rho_R)\theta_\pi} \left( \frac{Y_t}{Y_{t-1}} \right)^{(1 - \rho_R)\theta_Y} \epsilon_t^R.$$

The coefficient  $\bar{R}^b \geq 0$  determines the nominal interest rate in the steady state. The coefficients  $\theta_\pi, \theta_Y \geq 0$  govern the extent to which the central bank attempts to stabilize inflation and output growth,  $Y_t/Y_{t-1}$ . The parameter  $\rho_R \geq 0$  captures interest rate smoothing.



Government debt evolves according to the rule (see Woodford 1995)

$$(33) \quad \frac{B_{t+1}}{B_t} = \left(\frac{B_t}{\bar{B}}\right)^{-\gamma_B} \left(\frac{\pi_t}{\bar{\pi}}\right)^{\gamma_\pi} \left(\frac{Y_t}{\bar{Y}_{t-1}}\right)^{\gamma_Y} D_t, \quad D_t = D_{t-1}^{\rho_D} \epsilon_t^D,$$

where  $D_t$  is a persistent shock to the government's structural deficit. Besides issuing bonds, the government uses tax revenues  $T_t$ , defined below, to finance government consumption,  $G_t$ , and interest on debt. The parameters  $\gamma_B$ ,  $\gamma_\pi$ , and  $\gamma_Y$  measure, respectively, how the deficit reacts to outstanding debt, inflation, and the output gap.

The average tax rate in the economy is set according to a similar rule,

$$(34) \quad \frac{\tau_t}{\bar{\tau}} = \left(\frac{\tau_{t-1}}{\bar{\tau}}\right)^{\rho_\tau} \left(\frac{B_t}{B_{t-1}}\right)^{(1-\rho_\tau)\gamma_B^{\bar{\tau}}} \left(\frac{Y_t}{Y_{t-1}}\right)^{(1-\rho_\tau)\gamma_Y^{\bar{\tau}}}.$$

The level parameter of the tax code  $\tau_t^L$  adjusts such that the average tax rate on income equals this target level; i.e.,

$$(35) \quad \tau_t = \frac{E_t[w_t n_{it} h_{it} + \mathbf{1}\{h_{it} = 0\} \Pi_t^E] - \tau_t^L E_t[w_t n_{it} h_{it} + \mathbf{1}\{h_{it} = 0\} \Pi_t^E]^{\bar{\tau}^p}}{E_t[w_t n_{it} h_{it} + \mathbf{1}\{h_{it} = 0\} \Pi_t^E]},$$

where  $E_t$  is the expectation operator, which here gives the cross-sectional average. Total government tax revenues  $T_t$  are then  $T_t = \tau_t(w_t n_{it} h_{it} + \mathbf{1}\{h_{it} \neq 0\} \Pi_t^U + \mathbf{1}\{h_{it} = 0\} \Pi_t^E)$ , and the government budget constraint determines government spending residually:  $G_t = B_{t+1} + T_t - R_t^b/\pi_t B_t$ .

There are thus two shocks to government rules: monetary policy shocks,  $\epsilon_t^R$ , and structural deficit shocks,  $\epsilon_t^D$ . We assume these shocks to be log normally distributed with mean zero.

#### D. Goods, Asset, and Labor Market Clearing

The labor market clears at the competitive wage given in (4). The liquid asset market clears whenever the following equation holds:

$$(36) \quad B_{t+1} + q_t^\Pi = B^d(A_t, w_t, w_t^F, \Pi_t^E, \Pi_t^U, q_t, r_t, q_t^\Pi, q_{t-1}^\Pi, R_t^b, \pi_t, \pi_t^W, \tau_t, \Theta_t, \mathbb{W}_{t+1}) \\ := E_t[\lambda b_{a,t}^* + (1 - \lambda) b_{n,t}^*],$$

where  $b_{a,t}^*, b_{n,t}^*$  are functions of the states  $(b, k, h)$  and depend on how households value asset holdings in the future,  $\mathbb{W}_{t+1}(b, k, h)$ , and the current set of prices (and tax rates)  $(A_t, w_t, w_t^F, \Pi_t^E, \Pi_t^U, q_t, r_t, q_t^\Pi, q_{t-1}^\Pi, R_t^b, \pi_t, \pi_t^W, \tau_t, \Theta_t, \mathbb{W}_{t+1})$ . Future prices do not show up because we can express the value functions such that they summarize all relevant information on the expected future price paths. Expectations in the right-hand-side expression are taken with respect to the distribution  $\Theta_t(b, k, h)$ . Equilibrium requires the total *net* amount of bonds the household sector demands,  $B_t^d$ , to equal the supply of government bonds plus the value of profit shares. In gross terms there are more liquid assets in circulation; some households borrow up to  $\underline{B}$ .

The value of profit shares is, given the linearized solution, determined by a no-arbitrage condition between bonds and profit shares. Both need to have the same expected return:

$$(37) \quad q_t^\Pi R_t^b = E_t \pi_{t+1} \left[ (1 - \iota^\Pi) q_{t+1}^\Pi + \omega^\Pi \Pi_{t+1}^F \right].$$

Last, the market for capital has to clear; i.e.,

$$(38) \quad K_{t+1} = K^d(A_t, w_t, w_t^F, \Pi_t^E, \Pi_t^U, q_t, r_t, q_t^\Pi, q_{t-1}^\Pi, R_t^b, \pi_t, \pi_t^W, \tau_t, \Theta_t, \mathbb{W}_{t+1}) \\ := E_t [\lambda k_t^* + (1 - \lambda)k],$$

where  $K^d$  defines the aggregate supply of funds from households—both those that trade capital,  $\lambda k_t^*$ , and those that do not,  $(1 - \lambda)k$ . Again  $k_t^*$  is a function of the current prices and continuation values. The goods market then clears due to Walras's law, whenever labor, bonds, and capital markets clear.

When we consider the representative household model, we can think of  $K^d$  and  $B^d$  as simply given by (30) and (31). In other words, the representative household model only changes equilibrium conditions in replacing the Bellman equation and the capital and bonds demand equations but leaves the entire other model structure unchanged.

### E. Equilibrium

A *sequential equilibrium with recursive planning* in our model is a sequence of policy functions  $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$ ; a sequence of value functions  $\{\mathbb{W}_t\}$ ; a sequence of prices  $\{w_t, w_t^F, \Pi_t^E, \Pi_t^U, q_t, r_t, q_t^\Pi, R_t^b, \pi_t, \pi_t^W, \tau_t\}$ ; a sequence of stochastic states  $\{A_t, Z_t, \Psi_t, \mu_t^Y, \mu_t^W, D_t\}$  and shocks  $\{\epsilon_t^A, \epsilon_t^Z, \epsilon_t^\Psi, \epsilon_t^{\mu^Y}, \epsilon_t^{\mu^W}, \epsilon_t^D, \epsilon_t^R\}$ ; aggregate capital and labor supplies  $\{K_t, N_t\}$ ; distributions  $\Theta_t$  over individual asset holdings and productivity; and expectations for the distribution of future prices, such that

- (i) Given the functionals  $E_t \mathbb{W}_{t+1}$  for the continuation value and period- $t$  prices, policy functions  $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$  solve the households' planning problem, and given the policy functions  $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$  and prices, the value functions  $\{\mathbb{W}_t\}$  are a solution to the Bellman equation (27).
- (ii) Distributions of wealth and income evolve according to households' policy functions.
- (iii) The labor, the final goods, the bond, the capital, and the intermediate goods markets clear in every period; interest rates on bonds are set according to the central bank's Taylor rule; fiscal policies are set according to the fiscal rules; and stochastic processes evolve according to their law of motion.
- (iv) Expectations are model consistent.

## II. Numerical Solution and Estimation Technique

Solving for the sequential equilibrium of the model is a challenging task, in particular in an estimation setting. In the following, we develop our new Bayesian reduction technique starting from well-known principles in solving for state-space solutions of difference equations. First, we summarize how one can write a heterogeneous agent model in form of a system of difference equations, following Reiter (2009). Then, we explain how our method achieves the necessary complexity reduction that renders the solution algorithm fast enough to repeatedly obtain a state-space solution within the estimation procedure, even when the household problem is high dimensional. Third, we propose a metric to assess the quality of our solution relative to alternatives. Finally, we discuss how the proposed state-space solution is integrated in a Bayesian estimation.

### A. From Representative Agent to Heterogeneous Agent Solution

To understand the solution procedure we propose, it is useful to start by thinking about the representative agent twin of our model as in Section IB. Following Klein (2000) and Schmitt-Grohé and Uribe (2004), we represent the sequential equilibrium as the solution of a nonlinear difference equation

$$(39) \quad E_t F(X_t^*, X_{t+1}^*) = 0,$$

where  $X_t^*$  are combined state and control variables. While the partitioning of  $X_t^*$  into states and controls matters for the solution of (39), it is not central to our argument. Instead, in moving from the representative agent to the heterogeneous agent model, it is useful to partition the variables in  $X_t^*$  in terms of whether they are household choices or not. This means we separate out composite consumption,  $x_t$ , capital demand,  $K_t^d$ , and bond demand,  $B_t^d$ , and capture by  $X_t$  all other aggregate variables. Both  $B_t^d$  and  $K_t^d$  are state variables, and  $x_t$  is a control. The corresponding equilibrium conditions are the consumption Euler equations and the budget constraint of the household, equations (29)–(31). In equilibrium, the capital demanded by firms,  $K_t$ , and the bonds offered by the government,  $B_t$ , need to align with the household plans; i.e., (36) and (38) need to hold. This means we can write the representative household equilibrium as

$$(40) \quad E_t F(\{x_t, K_t^d, B_t^d\}, X_t, \{x_{t+1}, K_{t+1}^d, B_{t+1}^d\}, X_{t+1}) = 0.$$

Moving to the heterogeneous agent case, we simply replace the household variables by a large vector of variables and the corresponding optimality and equilibrium conditions (see Reiter 2009), discretizing the idiosyncratic state space. Specifically, the Bellman equation (27) pins down the optimal policy at each node  $(b, k, h)$  and the sequence of continuation values,  $\mathbb{W}_t(b, k, h)$ .<sup>15</sup> Similarly, for each node, (28) describes how the mass of households evolves over time. Taken together, (27) and

<sup>15</sup>In practice, our algorithm is based on the endogenous grid method of Carroll (2006) and Hintermaier and Koeniger (2010) and thus works with marginal value functions instead of value functions.

(28) define a set of equilibrium conditions for the evolution of continuation values and distribution masses—two conditions for each node  $(b, k, h)$ . These “replace” (29), (30), and (31) of the representative agent model. We summarize these “idiosyncratic variables,” continuation values, and masses by  $f_t$  in the following and obtain as the equilibrium condition, with a slight abuse of notation,

$$(41) \quad E_t F(f_t, X_t, f_{t+1}, X_{t+1}) = 0.$$

In principle, a first-order approximation of (41) can be solved in the usual way: calculating the Jacobian of  $F$  and, e.g., running a QZ-decomposition as proposed by Klein (2000).

### B. Reducing Complexity

In practice, however, the direct attack to linearizing and solving (41) becomes infeasible, given the size of the system in our application. This is due to the high number of nodes of the liquid/illiquid asset-income grid ( $100 \times 100 \times 22$ ). Reiter (2009); Bayer and Luetticke (2020); and Ahn et al. (2018) suggest methods that tackle the issue based on reducing the number of state and control variables. The three approaches do so, respectively, before solving for the stationary equilibrium, after solving for this equilibrium but before knowledge of the dynamics, or after linearizing the system of differential equations. These approaches render a *single solution* of the economic model at a good quality of approximation numerically feasible. For *estimation*, we develop a novel Bayesian reduction technique for HANK models that is more aggressive and also better informed. Below, we provide an overview; further technical details can be found in online Appendix C.

*Using the Partitioning of Variables and Parameters to Simplify the Linearization.*—The first improvement is based on the observation that the dimensionality of the heterogeneous agent part,  $f_t$ , is much larger than that of the aggregate part,  $X_t$ . Keeping this partitioning, we write the linearized version of the difference equation (41) as

$$(42) \quad \underbrace{\begin{bmatrix} B_{ff} & B_{fX} \\ B_{Xf} & B_{XX} \end{bmatrix}}_{=B} \begin{bmatrix} f_t \\ X_t \end{bmatrix} = -E_t \underbrace{\begin{bmatrix} A_{ff} & A_{fX} \\ A_{Xf} & A_{XX} \end{bmatrix}}_{=A} \begin{bmatrix} f_{t+1} \\ X_{t+1} \end{bmatrix},$$

where we ordered, in line with the ordering of variables, the set of “idiosyncratic” equations—(27) and (28)—first and all other equations last.

This partitioning is useful not only conceptually but also very practically. The idiosyncratic equations contain only a small subset of the parameters of the model. In other words,  $A_{ff}$ ,  $B_{ff}$ ,  $A_{fX}$ , and  $B_{fX}$  only depend on this small subset. What is more,  $A_{Xf}$  and  $B_{Xf}$  can be made parameter free if we introduce the aggregate capital/bond holdings of households as auxiliary variables such that the distribution no longer enters the aggregate model block directly. Therefore, we only need to update  $B_{XX}$  and  $A_{XX}$  when changing parameters during the estimation that do not directly appear in the household problem. For example, changing the fiscal-policy or Taylor rule

coefficients or nominal rigidities only affects  $B_{XX}$  and  $A_{XX}$ . This means that the number of derivatives to be calculated in the second step, i.e., when estimating the business cycle model with heterogeneous agents, is the same as the number of derivatives to be updated during the estimation of its representative agent “twin.”

Importantly, those parameters directly affecting the other “idiosyncratic” blocks also affect the stationary equilibrium. This means they can be identified (calibrated) in an incomplete markets model using time series averages of aggregate and cross-sectional data, approaching the estimation effectively in two steps. In our case these include the discount factor, the liquidity of assets, borrowing constraints, and the average income risk.

*Dimensionality Reduction Prior to Knowledge of the Dynamics of the Economy.*—Naturally, the partitioning itself does not solve the problem of  $A$  and  $B$  being very large matrices because of the high-dimensional idiosyncratic grid. To tackle this problem, we proceed in two steps. First, we reduce the dimensionality without knowledge of the dynamics of the economy (where we extend Bayer and Luetticke 2020). Second, we leverage that, after solving the model once, we can reduce the model further based on the dynamics of the model.

The first step is to think of  $f_t$  as parameters of the deviation of value functions and distributions from their stationary equilibrium counterparts. This view of  $f_t$  as parameters of functions immediately implies that the dimensionality of  $f_t$  no longer needs to be tightly connected to the size of the asset-income grid. We construct these difference functions as a mix of their representation<sup>16</sup> by a multidimensional discrete cosine transform (DCT) and linear interpolations. The multidimensional DCT can be expressed in terms of Chebychev polynomials evaluated only at the nodal values of the marginal value functions  $\partial \mathbb{W}_t / \partial b$  and  $\partial \mathbb{W}_t / \partial k$  used in the first-order conditions of the optimization problem (27), where the assumption is that these nodal values are Chebychev nodes. When perturbing the problem (27), we perturb the coefficients instead of the function values themselves. To reduce complexity, we treat those coefficients as fixed (and do not perturb them) that are numerically small (in absolute value) in the stationary equilibrium or do not affect the partial derivative of the marginal value functions with respect to prices. The latter aspect improves on Bayer and Luetticke (2020); see online Appendix C.1. The multidimensional DCT and its inverse can be used to efficiently transform nodal values to coefficients and reverse. We also use the DCT to project the equilibrium conditions to the space of the perturbed coefficients.

The distribution function  $\Theta_t$ , see (28), we represent by marginals,  $F_t^b, F_t^k, F_t^h$ , and a copula,  $C_t(\cdot)$ , following Bayer and Luetticke (2020).<sup>17</sup> Improving on their approach, however, we write the copula  $C_t(F_t^b, F_t^k, F_t^h)$  at time  $t$  as the sum of the linear interpolants generated from the steady-state copula  $\bar{C}(\cdot)$  and a new perturbation term,  $\hat{C}_t(\cdot)$ . This perturbation term is a linear interpolant, too, but we allow it to have a different and sparser nodal grid, compared to the first term. The marginal

<sup>16</sup>To reduce the curvature of the marginal value functions, we first transform them by the inverse marginal utility and then take the logarithm.

<sup>17</sup>There is a numerical advantage of writing the distribution in the form of a copula and marginals. The copula reflects cross-terms, which can be expected to be of lesser importance for aggregate dynamics. The separation then allows us to calibrate separate degrees of precision in line with the differences in importance.

distributions,  $F_t^b, F_t^k, F_t^h$ , enter directly in  $f_t$ . More details can be found in online Appendix C.1.

*Dimensionality Reduction Based on Knowledge of the Economy's Dynamics.*—

This first reduction step allows us to solve the model, but if we want the solution to be sufficiently precise, the number of variables in the difference equation remains high, and thus the computational time for the QZ-decomposition long. One way to address this issue is further model reduction. Ahn et al. (2018) give an overview of model-reduction techniques for difference equations. Model reduction in our case means finding an orthonormal basis  $\mathcal{P} \in \mathbb{R}^{n \times m}$  with  $m \ll n$  such that we can write  $f_t \approx \mathcal{P}Y_t$  and replace the system (42) by a system with factors  $Y_t$ ,

$$(43) \quad \underbrace{\begin{bmatrix} \mathcal{P}'B_{ff}\mathcal{P} & \mathcal{P}'B_{fx} \\ B_{xf}\mathcal{P} & B_{xx} \end{bmatrix}}_{=B'} \begin{bmatrix} Y_t \\ X_t \end{bmatrix} = -E_t \underbrace{\begin{bmatrix} \mathcal{P}'A_{ff}\mathcal{P} & \mathcal{P}'A_{fx} \\ A_{xf}\mathcal{P} & A_{xx} \end{bmatrix}}_{=A'} \begin{bmatrix} Y_{t+1} \\ X_{t+1} \end{bmatrix}.$$

The solution of this reduced model should give us an arbitrarily close approximation to the solution of the original problem (42).

For a given set of parameters, this can be achieved by solving first the original system (42) given a prior parameterization that includes the shock processes. Based on this solution, we calculate the variance-covariance matrix,  $\Sigma_f$ , of each subgroup of elements in  $f$  (the value functions and the copula separately) and perform a Jordan eigenvalue decomposition thereof (and thus obtain their principal components). This gives us a factorization of  $f$ , where factors associated with small eigenvalues,  $\lambda_{2f}$ , are approximately constant and thus irrelevant for the model dynamics and in particular the estimation

$$(44) \quad \Sigma_f = \begin{bmatrix} \mathcal{Q}'_{1f} & \mathcal{Q}'_{2f} \end{bmatrix} \begin{bmatrix} \lambda_{1f} & 0 \\ 0 & \lambda_{2f} \end{bmatrix} \begin{bmatrix} \mathcal{Q}_{1f} \\ \mathcal{Q}_{2f} \end{bmatrix}.$$

This means we can choose  $\mathcal{P} = \mathcal{Q}'_{1f}$  as the basis for the model reduction. The reduced model then, by construction, has a tiny approximation error relative to the full model at the parameterization used to construct  $\mathcal{P}$ . Furthermore, one can expect that this basis  $\mathcal{P}$  remains a reasonably good basis in the vicinity of the parameters.<sup>18</sup> Therefore, our algorithm updates  $\mathcal{P}$  only infrequently. We show, for our application, that, even when parameters change but  $\mathcal{P}$  does not, the solution of the reduced model is a precise approximation of the full model solution. In online Appendix C.2 and C.3, we argue in more detail why one can expect a strong model reduction possibility and why this reduction can be expected to be stable to parameter variations in the estimation.

In practice, we generate  $\mathcal{P}$  initially based on the model's prior mode and then update it once during posterior-mode finding and once after finding a tentative posterior mode of the parameter distributions, but before running the Markov chain Monte Carlo algorithm. In our application, the model reduction leaves us with 504

<sup>18</sup> Ahn et al. (2018) argue that small variations compared to the ideal minimal basis can be compensated by retaining a somewhat too high dimensional basis.

state variables and 70 controls. This includes 21 aggregate state variables and 44 aggregate controls.

### C. Assessment of Quality

We check the precision of our second model-reduction step by comparing the likelihood and the impulse response functions (IRF) across model solutions. The first solution we compare to is the one in (42) without the second model-reduction step. The second solution we compare to is the sequence-space method proposed by Boppart, Krusell, and Mitman (2018) and Auclert et al. (2021).

We suggest to evaluate the distance between two model solutions based on the IRFs they produce for the data used in estimation. Concretely, we define the distance between the reference model solution  $S1$  and the alternative solution  $S2$  as

$$(45) \quad D_{S1,S2}(H,x|\vartheta) = 1 - \frac{\sum_{s \in S} \sum_{h=1}^H [IRF_{S1}(x,s,h|\vartheta) - IRF_{S2}(x,s,h|\vartheta)]^2}{\sum_{s \in S} \sum_{h=1}^H [IRF_{S1}(x,s,h|\vartheta)]^2},$$

where  $H$  is the horizon up to which we evaluate the response of variable  $x$  to a one standard deviation shock  $\epsilon_s$ , summing over all aggregate shocks,  $s \in S$ .  $\vartheta$  is the parameter vector. The metric is akin to a forecast error variance decomposition: the distance informs us about the fraction of the forecast error variance for  $x$  under the reference solution  $S1$  that is “explained” by the alternative solution  $S2$ . We show in Section IIID that our solution is practically identical to both reference solutions.

### D. Estimation

As explained, it is useful to group the model parameters  $\vartheta$  into those that affect the stationary equilibrium and those that do not when thinking about estimating the model. The representative agent literature knows this split, too, and sometimes calibrates parameters like steady-state markups or average government expenditure shares (e.g., Smets and Wouters 2007; Christiano, Eichenbaum, and Evans 2005). In the heterogeneous agent setting, the set of parameters that affect the stationary equilibrium is larger. Thus, more parameters can be calibrated. At the same time, parameters that affect the steady state are more costly to estimate from the time series properties of data because they require recalculating the stationary equilibrium. For this reason, we calibrate those parameters to cross-sectional moments and time series averages of the data. This means that in our application, compared to the typical representative agent estimation that tries to calibrate as little as possible (e.g., Justiniano, Primiceri, and Tambalotti 2011), we additionally include the discount factor, the risk aversion, the Frisch elasticity of labor supply, and the capital share in the set of calibrated parameters. Of course, we also calibrate the forcing processes of heterogeneity itself. We come back to this in more detail in Section III.

Those parameters that do not affect the stationary equilibrium can be estimated from the time series evolution of the data (cross-sectional and aggregate). For this estimation, we use an off-the-shelf Bayesian approach as described in An and Schorfheide (2007) and Fernández-Villaverde (2010). In particular, we use the



Kalman filter to obtain the likelihood from the state-space representation of the model solution and employ a standard random walk Metropolis-Hastings (RWMH) algorithm to generate draws from the posterior likelihood after an extensive mode finding. Smoothed estimates of the states at the posterior mean of the parameters are obtained via a Kalman smoother of the type described in Koopman and Durbin (2000) and Durbin and Koopman (2012). One likelihood evaluation takes circa 750 ms on a desktop computer (Intel i7-10700K, code written in Julia); 80 percent of the time is needed for the model solution, the remainder for the Kalman filter.<sup>19</sup> The full grid for income and wealth has 220,000 nodes, and the model after the first step of dimensionality reduction that is used to calculate  $\mathcal{P}$  has 2,883 variables. Finding the stationary equilibrium takes roughly 7 minutes; the first linearized solution that does not yet use the second model reduction takes less than 15 minutes to compute.

We deliberately choose the most commonly used approach in estimating DSGE models, but our state-space-based solution approach also lends itself directly to more sophisticated estimation techniques. One particular example is the Sequential Monte Carlo approach (Herbst and Schorfheide 2014, 2015) that Acharya et al. (2021) advocate for the estimation of large heterogeneous agent business cycle models.

### III. Parameterization, Priors, Posteriors, and Quality

We estimate several variants of the model using the procedure outlined above. In the following, we focus on two main variants and compare them to their representative agent twin; details on all other variants can be found in the online Appendix. First, we estimate the model on aggregate data alone and allow only for standard aggregate shocks (HANK). Second, we include in the estimation distributional time series data and allow for shocks to income risk and tax progressivity, for which we also add observables (HANK-X).<sup>20</sup>

#### A. Calibrated Parameters

We fix some parameters that affect the stationary equilibrium targeting average data ratios; see Table 1 (all at the quarterly frequency of the model).<sup>21</sup> In addition, we directly take some parameter estimates from the literature. In particular, we take the idiosyncratic income process from Storesletten, Telmer, and Yaron (2004), which gives us  $\rho_h = 0.98$  and  $\bar{\sigma}_h = 0.12$ . Guvenen, Kaplan, and Song (2014) provide the probability that a household will fall out of the top 1 percent of the income distribution in a given year, which we take as the transition probability from entrepreneur to worker,  $\iota = 6.25\%$ . We set the relative risk aversion,  $\xi$ , to 4, which is common in the incomplete markets literature; see Kaplan and Violante (2014).<sup>22</sup>

<sup>19</sup>We do not leverage the fact that the model solution can be obtained without repeating the QZ-decomposition when changing only the parameters of the forcing processes. When only parameters of the exogenous shock processes are estimated, the model solution can, as usual, be obtained in virtually no time.

<sup>20</sup>See footnotes 12 and 13 for the adjustments to the baseline model.

<sup>21</sup>Online Appendix A.2 provides the table of steady-state parameters for the recalibrated representative agent analog of our model.

<sup>22</sup>We also estimate our model with a relative risk aversion of 2; see Section IVC and online Appendix B.

TABLE 1—CALIBRATION (*quarterly frequency*)

| Par.                                   | Value  | Description                  | Target                                                 | Data source                |
|----------------------------------------|--------|------------------------------|--------------------------------------------------------|----------------------------|
| <i>Households: income process</i>      |        |                              |                                                        |                            |
| $\rho_h$                               | 0.980  | Persistence labor income     | Storesletten, Telmer, and Yaron (2004)                 |                            |
| $\sigma_h$                             | 0.120  | Std. labor income            | Storesletten, Telmer, and Yaron (2004)                 |                            |
| $\iota$                                | 0.063  | Trans. prob. E. to W.        | Güvenen, Kaplan, and Song (2014)                       |                            |
| $\zeta$                                | 2.0E-5 | Trans. prob. W. to E.        | Top 10 wealth share: 67 percent                        | WID 1954–2019              |
| <i>Households: financial frictions</i> |        |                              |                                                        |                            |
| $\lambda$                              | 0.062  | Portfolio adj. prob.         | Liquid to illiquid, $\frac{B + \bar{q}^\Pi}{K} = 0.25$ | SCF 1950–2016 <sup>a</sup> |
| $\bar{R}$                              | 0.021  | Borrowing penalty            | Share of borrowers: 16 percent                         | SCF 1983–2016 <sup>b</sup> |
| $\bar{q}^\Pi/Y$                        | 1.140  | Value of profit shares       | Gov. debt to output, $\frac{B}{Y} = 1.67$              | NIPA 1954–2019             |
| <i>Households: preferences</i>         |        |                              |                                                        |                            |
| $\beta$                                | 0.983  | Discount factor              | Capital to output, $\frac{K}{Y} = 11.22$               | NIPA 1954–2019             |
| $\xi$                                  | 4.000  | Relative risk aversion       | Kaplan and Violante (2014)                             |                            |
| $\gamma$                               | 2.000  | Inverse of Frisch elasticity | Chetty et al. (2011)                                   |                            |
| <i>Firms</i>                           |        |                              |                                                        |                            |
| $\alpha$                               | 0.680  | Share of labor               | Average labor income share                             | Standard value             |
| $\delta_0$                             | 0.018  | Depreciation rate            | Bayer, Born, and Luetticke (2023)                      |                            |
| $\bar{\eta}$                           | 11.000 | Elasticity of substitution   | Born and Pfeifer (2014)                                |                            |
| $\bar{\zeta}$                          | 11.000 | Elasticity of substitution   | Born and Pfeifer (2014)                                |                            |
| <i>Government</i>                      |        |                              |                                                        |                            |
| $\bar{\tau}^L$                         | 0.180  | Tax rate level               | Gov. consumption, $\frac{G}{Y} = 0.2$                  | NIPA 1954–2019             |
| $\bar{\tau}^P$                         | 0.102  | Tax progressivity            | Average progressivity                                  | SOI 1954–2019              |
| $\bar{R}^b$                            | 1.000  | Gross nominal rate           | Growth $\approx$ interest rate                         |                            |
| $\bar{\pi}$                            | 1.000  | Gross inflation              | Indexation, w.l.o.g.                                   |                            |

Notes: Calibration targets are the sample averages when a data source is given. Otherwise, the parameter is fixed to the value in the cited literature. BLS: Bureau of Labor Statistics. NIPA: National Income and Product Accounts. SCF: Survey of Consumer Finances; shares are taken from <sup>a</sup>Weiß (2023, Table 2.1, p. 144) and <sup>b</sup>Bayer et al. (2019, Table 3, p. 277). SOI: Statistics of Income. WID: World Inequality Database. Details on the data can be found in online Appendix A.

We set the Frisch elasticity to 0.5; see Chetty et al. (2011). The steady-state price and wage markups are both fixed to 10 percent, following Born and Pfeifer (2014).

All other calibrated parameters are closely tied to time series averages of data moments. While they are calibrated jointly, we present them as if they are only informed by the statistic that is most informative for a given parameter. The transition probability to become an entrepreneur,  $\zeta = 0.0002$ , pins down the per capita profits of entrepreneurs and therefore the top ten share in wealth. The discount factor,  $\beta = 0.983$ , and the liquidity of assets,  $\lambda = 0.062$ , pin down the capital to output ratio and the share of liquid assets in household portfolios. The borrowing penalty  $\bar{R} = 2.06\%$  determines how many households are indebted. The difference between total liquidity and government bonds pins down the value of profit shares relative to output,  $\bar{q}^\Pi/Y = 1.14$ , which determines the ratio of  $\omega^\Pi$  and  $\iota^\Pi$ , as  $\omega^\Pi/\iota^\Pi = \eta\bar{q}^\Pi/Y$  through the steady-state version of (37). We set steady-state inflation to zero, as we have assumed indexation to the steady-state inflation rate in the Phillips curves. We set the steady-state net interest rate on bonds also to zero, in

order to broadly capture the average federal funds rate in real terms minus output growth over 1954 to 2019.

The share of labor in production,  $\alpha = 0.68$ , is pinned down by the average labor income share (given  $\eta$ ). The average quarterly depreciation,  $\delta_0 = 0.0175$ , can be read off the average depreciation rates on US capital (including buildings). We set the average taxation level,  $\bar{\tau}^L = 0.18$ , such that the budget balances for the observed level of government consumption. Finally, we follow Ferriere and Navarro (forthcoming) in constructing a direct estimate for tax progressivity, extending their estimates until 2017; see online Appendix A.1.2. This approach uses the assumed nonlinear tax schedule to measure progressivity and is based on Mertens and Montiel Olea's (2018) estimates of average marginal tax rates. Heathcote, Storesletten, and Violante (2017) show that this functional form approximates the progressivity of the US tax system well. The tax progressivity exponent,  $\bar{\tau}^P = 0.1022$ , matches the time series average of this statistic.

### *B. Time Series Data Used for Estimation*

For the estimation, we use quarterly US data from 1954:III to 2019:IV and include the following seven observable time series: the growth rates of per capita GDP, private consumption, investment, and wages, all in real terms; the logarithm of the level of per capita hours worked; the log difference of the GDP deflator; and the (shadow) federal funds rate. Our model is stationary, so all growth rates are de-measured; see online Appendix A.1.2 for a formal depiction of the vector of observables. These data are standard in the estimation of DSGE models.

In the HANK-X extension, we add more data with shorter and/or nonquarterly availability. First, cross-sectional information on wealth and income shares of the top 10 percent. These are available at an annual frequency from 1954 to 2019 from the World Inequality Database.<sup>23</sup> The reason we focus on the top ten wealth and income shares is that these measures are most similar across alternative, but less frequently available, data sources, such as the Survey of Consumer Finances; see Kopczuk (2015). Second, we use the time series of the tax progressivity estimates that we construct in online Appendix A.1.2. Third, we add income risk estimates, available at a quarterly frequency from 1983:I to 2013:I, from Bayer et al. (2019) based on panel data in the Survey of Income and Program Participation. In this extended estimation we also allow for shocks to progressivity and income risk, which directly affect the distribution of income. We allow for measurement error on the cross-sectional data (the top ten shares) to avoid stochastic singularity. For simplicity, we treat the measurement error on the top ten shares as classical (normal and i.i.d.) despite the fact that the shares are nonlinear functions of the sampled micro data. In other words, we assume that measurement errors reflect more than just sampling uncertainty.

<sup>23</sup>This database draws on work by Piketty, Saez, and Zucman; see, e.g., Piketty and Saez (2003) or Saez and Zucman (2016).

### C. Priors and Posteriors

Columns 1–4 of Table 2 present the parameters we estimate, their assumed prior distributions, and their posteriors. The priors and posteriors for the distributions of the shock processes are listed in online Appendix A.3. Where available, we use prior values that are standard in the literature and independent of the underlying data.

*Priors.*—Following Justiniano, Primiceri, and Tambalotti (2011), we impose a gamma distribution with prior mean of 5.0 and standard deviation of 2.0 for  $\delta_2/\delta_1$ , the elasticity of marginal depreciation with respect to capacity utilization, and a gamma prior with mean 4.0 and standard deviation of 2.0 for the parameter controlling investment adjustment costs,  $\phi$ . For the slopes of price and wage Phillips curves,  $\kappa_Y$  and  $\kappa_w$ , we assume gamma priors with mean 0.10 and standard deviation 0.03. This corresponds to price and wage contracts having an average length of four quarters at the prior mode. Regarding the profit-shares parameters  $\iota^\Pi$  and  $\omega^\Pi$ , we assume that  $\iota^\Pi$  follows a shifted beta distribution with mean 0.5 and standard deviation 0.25. We set the extrema of the shifted distribution such that the expected duration of the profit shares,  $1/\iota^\Pi$ , is at least 10 years and at most 200 years. The duration at the prior mode is 20 years. The value for  $\omega^\Pi$  then follows from keeping the stationary equilibrium value of  $\bar{q}^\Pi$  constant.

For monetary policy, we estimate feedback parameters in the Taylor rule for inflation and output growth,  $\theta_\pi$  and  $\theta_Y$ . We impose normal distributions with prior means of 1.7 and 0.13, respectively. In addition, we allow for interest rate smoothing with parameter  $\rho_R$ . Here, we assume a beta distribution with parameters (0.5, 0.2).

In the bond rule, the debt-feedback parameter  $\gamma_B$  is assumed to follow a gamma distribution with mean 0.10 and standard deviation 0.08. This centers the prior for the autocorrelation of debt around 0.9 and implies a half-life of between one and eight years for a deviation in debt. The parameters governing the feedback to inflation and output growth,  $\gamma_\pi$  and  $\gamma_Y$ , follow standard normal distributions. Similarly, the autoregressive parameters, in the tax rules,  $\rho_P$  and  $\rho_\tau$ , are assumed to follow beta distributions (with mean 0.5 and standard deviation 0.2). The feedback parameters for average tax rates,  $\gamma_Y^\tau$  and  $\gamma_B^\tau$ , follow standard normal distributions.

Following Smets and Wouters (2007), the autoregressive parameters of the shock processes are assumed to follow a beta distribution with mean 0.5 and standard deviation 0.2. The standard deviations of the shocks follow inverse-gamma distributions with prior mean 0.1 percent and standard deviation 2 percent.<sup>24</sup> In our baseline we do not include measurement errors but allow for these when including estimates of top income and wealth shares as additional data.

*Posteriors.*—In Table 2, columns 5–7 report the posterior distributions across the three main estimation variants: RANK, HANK, and HANK-X. Here, we focus on the frictions and policy parameters. The estimated parameters of the exogenous driving processes can be found in online Appendix A.3. Checks on the convergence

<sup>24</sup>In the HANK-X extension, we use a higher prior mean for income risk shocks,  $s$ , given the evidence in Bayer et al. (2019).

TABLE 2—PRIOR AND POSTERIOR DISTRIBUTIONS OF ESTIMATED PARAMETERS

| Parameter                                             | Distribution | Prior |        | Posterior                  |                            |                            |
|-------------------------------------------------------|--------------|-------|--------|----------------------------|----------------------------|----------------------------|
|                                                       |              | Mean  | SD     | RANK                       | HANK                       | HANK-X                     |
| <i>Frictions</i>                                      |              |       |        |                            |                            |                            |
| $\delta_s$                                            | Gamma        | 5.00  | 2.00   | 0.697<br>(0.405, 1.058)    | 1.062<br>(0.816, 1.327)    | 0.706<br>(0.532, 0.900)    |
| $\phi$                                                | Gamma        | 4.00  | 2.00   | 7.875<br>(5.989, 10.033)   | 3.790<br>(3.104, 4.555)    | 1.958<br>(1.826, 2.090)    |
| $\kappa$                                              | Gamma        | 0.10  | 0.03   | 0.119<br>(0.088, 0.155)    | 0.168<br>(0.129, 0.211)    | 0.140<br>(0.108, 0.177)    |
| $\kappa_w$                                            | Gamma        | 0.10  | 0.03   | 0.282<br>(0.206, 0.364)    | 0.280<br>(0.216, 0.348)    | 0.238<br>(0.178, 0.303)    |
| $\iota^\Pi$                                           | Beta         | 0.50  | 0.25   | —<br>(—, —)                | 0.312<br>(0.049, 0.709)    | 0.660<br>(0.318, 0.928)    |
| <i>Monetary policy</i>                                |              |       |        |                            |                            |                            |
| $\rho_R$                                              | Beta         | 0.50  | 0.20   | 0.794<br>(0.768, 0.818)    | 0.787<br>(0.761, 0.812)    | 0.807<br>(0.783, 0.831)    |
| $\sigma_R$                                            | Inv.-Gamma   | 0.10  | 2.00   | 0.238<br>(0.218, 0.258)    | 0.233<br>(0.214, 0.253)    | 0.229<br>(0.210, 0.249)    |
| $\theta_\pi$                                          | Normal       | 1.70  | 0.30   | 2.162<br>(1.979, 2.361)    | 1.947<br>(1.742, 2.164)    | 2.066<br>(1.841, 2.311)    |
| $\theta_Y$                                            | Normal       | 0.13  | 0.05   | 0.254<br>(0.184, 0.323)    | 0.202<br>(0.135, 0.270)    | 0.218<br>(0.149, 0.287)    |
| <i>Fiscal policy: deficit</i>                         |              |       |        |                            |                            |                            |
| $\rho_D$                                              | Beta         | 0.50  | 0.20   | 0.960<br>(0.927, 0.983)    | 0.958<br>(0.921, 0.989)    | 0.965<br>(0.933, 0.991)    |
| $\gamma_B$                                            | Gamma        | 0.10  | 0.08   | 0.089<br>(0.039, 0.140)    | 0.042<br>(0.009, 0.081)    | 0.013<br>(0.002, 0.034)    |
| $\gamma_\pi$                                          | Normal       | 0.00  | 1.00   | −2.756<br>(−3.164, −2.373) | −2.898<br>(−3.26, −2.569)  | −2.159<br>(−2.364, −1.973) |
| $\gamma_Y$                                            | Normal       | 0.00  | 1.00   | −0.734<br>(−0.898, −0.574) | −0.794<br>(−0.888, −0.702) | −0.43<br>(−0.489, −0.376)  |
| <i>Fiscal policy: taxes</i>                           |              |       |        |                            |                            |                            |
| $\rho_\tau$                                           | Beta         | 0.50  | 0.20   | 0.492<br>(0.368, 0.603)    | 0.411<br>(0.245, 0.559)    | 0.496<br>(0.395, 0.595)    |
| $\gamma_B^\tau$                                       | Normal       | 0.00  | 1.00   | 3.442<br>(2.350, 4.608)    | 2.796<br>(1.820, 3.885)    | 3.289<br>(3.250, 3.332)    |
| $\gamma_Y^\tau$                                       | Normal       | 0.00  | 1.00   | −1.621<br>(−3.004, −0.231) | 1.483<br>(1.013, 1.900)    | −0.921<br>(−0.944, −0.90)  |
| <i>Income risk</i>                                    |              |       |        |                            |                            |                            |
| $\rho_s$                                              | Beta         | 0.70  | 0.20   | —<br>(—, —)                | —<br>(—, —)                | 0.518<br>(0.445, 0.583)    |
| $\Sigma_Y$                                            | Normal       | 0.00  | 100.00 | —<br>(—, —)                | —<br>(—, —)                | 28.878<br>(28.862, 28.894) |
| log marginal data density (only aggregate data)       |              |       |        | 6,590                      | 6,588                      |                            |
| log marginal data density (with cross-sectional data) |              |       |        |                            |                            | 6,627                      |

*Notes:* The table displays the estimated parameters, their priors, and posterior means across three model variants: HANK, HANK-X, and RANK. The parameters of the shock processes are shown in online Appendix A.3. The 90 percent credible intervals are shown in parentheses. Posteriors are obtained by a Markov chain Monte Carlo method. The standard deviations have been multiplied by 100 for better readability. The parameter actually estimated and displayed is  $(800\iota^\Pi - 1)/19$  to ensure an expected duration of the profit shares between 10 and 200 years.

of the estimator are provided in online Appendix A.8.<sup>25</sup> The parameter estimates for HANK and HANK-X are typically close to the RANK estimates with few notable differences.

In particular, the estimated investment friction and the price rigidities in RANK are a bit larger. For the investment adjustment costs, this reflects, in part, that the portfolio adjustment costs at the household level already generate inertia in aggregate investment. Our estimates of the wage- and price-setting frictions imply that wages adjust roughly every two to three quarters in all models, and prices adjust every third quarter in HANK and every fourth quarter in RANK.

The estimated policy rules are even more similar across models. There is substantial interest rate inertia. All variants estimate a coefficient of 0.8 for interest smoothing. The Taylor rule coefficient on inflation is between 2.0 and 2.2; the one on output growth is between 0.2 and 0.3. The fiscal rule that governs deficits and hence government spending exhibits a countercyclical response to inflation with elasticities between  $-2.9$  and  $-2.2$ . The elasticities with respect to output growth are between  $-0.8$  and  $-0.4$ . Deficits feature a high degree of persistence as well. The tax rule that governs average taxation has much less inertia. This implies, given the transitory nature of output growth variations, that tax rates respond mostly to the level of government debt. More debt implies higher taxes.

When we estimate the income risk process in HANK-X, we find income risks to be procyclical in the sense that they go up when other shocks drive up output growth. Given the size of shocks that we estimate, the feedback is small. A 1 percent output growth increase leads to a 0.55 percent higher income risk; a 1 standard deviation shock to income risk leads to a 70 percent higher level of risk.

The posterior for  $\iota^{\Pi}$  implies a duration of profit shares of 27 years on average in the HANK specification and 14 years in HANK-X. Using the top 10 income data leads to a lower duration and therefore higher payout share  $\omega^{\Pi}$  (11 percent versus 21 percent). The higher payout share implies a smoother income inequality series but amplifies the volatility of aggregate consumption. Adding the cross-sectional data substantially increases the precision of the estimate, i.e., shrinks the credible intervals.

Table 2 also reports the marginal data densities for the three model estimates. RANK and HANK frameworks fit the aggregate data equally well. What is more, the HANK models predict certain correlations of cross-sectional and aggregate data that RANK, by construction, cannot.<sup>26</sup>

#### D. *Quality of the Model Reduction*

We assess the quality of our model reduction based on the estimated posterior distribution. First, we evaluate the quality of the second stage of the model reduction. For this purpose, we draw 1,000 parameter vectors from the posterior

<sup>25</sup> We estimate each model using a single RWMH chain after an extensive mode search. After a long burn-in, 400,000 draws from the posterior are used to compute the posterior statistics. The acceptance rates across chains are between 20 percent and 30 percent. Online Appendix A.8 provides Geweke (1992) convergence statistics as well as trace plots of individual parameters.

<sup>26</sup> We use this cross-sectional data in the estimation of HANK-X, which also means we cannot directly compare its marginal data density with the two other estimates.

TABLE 3—IRF VARIATION CAPTURED BY STATE-SPACE SOLUTION  
WITH SECOND-STAGE MODEL REDUCTION

| Observable          | Relative to no<br>second stage |        | Relative to<br>sequence space |        |
|---------------------|--------------------------------|--------|-------------------------------|--------|
|                     | HANK                           | HANK-X | HANK                          | HANK-X |
| Output growth       | 100.00                         | 100.00 | 100.00                        | 100.00 |
| Investment growth   | 100.00                         | 100.00 | 100.00                        | 100.00 |
| Consumption growth  | 100.00                         | 100.00 | 99.95                         | 99.96  |
| Hours worked        | 100.00                         | 100.00 | 99.99                         | 99.98  |
| Wage growth         | 100.00                         | 100.00 | 100.00                        | 100.00 |
| Policy rate         | 100.00                         | 100.00 | 99.98                         | 99.98  |
| Inflation           | 100.00                         | 100.00 | 99.98                         | 99.98  |
| Top 10 wealth share | —                              | 100.00 | —                             | 99.95  |
| Top 10 income share | —                              | 100.00 | —                             | 99.99  |
| Tax progressivity   | —                              | 100.00 | —                             | 100.00 |
| Income risk         | —                              | 100.00 | —                             | 100.00 |

*Notes:* The table displays (in percent) variation of IRFs based on a baseline solution that is captured by our state-space solution with second-stage model reduction. This statistic measures the distance between impulse responses generated by solving the model with different solution techniques and is based on (45). Columns 2 and 3 compare the solution with and without second-stage model reduction, columns 4 and 5 our solution technique to a sequence-space Jacobian solution. Columns 2 and 3 refer to the first percentile obtained by drawing 1,000 parameter vectors from the posterior distribution. Columns 4 and 5 refer to the posterior mean. All columns report the minimum (i.e., largest distance) over the first 32 quarters. A value of exactly 100 means that impulse responses are identical.

distribution, and, for each parameter draw, we solve the model once with, and once without, the second-stage reduction. The second-stage reduction matrix  $\mathcal{P}$  is kept constant throughout the experiments. This gives us a sampled distribution of distances between the two model solutions based on the impulse response functions. We find that the distance between the two solutions is basically zero. Table 3 reports the first percentile of the sampled quality measure (i.e., the largest distance). Note that the first-stage reduction is, by construction, invariant to the estimated parameters. This means that, at least in the vicinity of the estimated parameters, there is virtually no loss from using the second-stage model reduction even when  $\mathcal{P}$  is not reoptimized.

Still, one might be concerned that the necessary first-stage reduction introduces approximation errors. To assess this, we evaluate the solution under our method against a sequence-space Jacobian solution. Auclert et al. (2021) argue that the approximation quality of such method is very good if the horizon is chosen sufficiently long. Also here, we find that the distance between the solution techniques in terms of IRFs is extremely small. Given that we have shown the invariance of the model reduction in our first experiment, we only calculate the distance at the posterior mean.

Concretely, we calculate for both sets of experiments our metric for all observable variables used in the estimation. The metric is calculated based on all shocks using their estimated variances. Columns 2 and 4 of Table 3 show the statistics for the HANK specification, columns 3 and 5 for HANK-X. Here, we report the largest distance over the first 32 quarters. The distances are, as mentioned



before, minimal.<sup>27</sup> In online Appendix C.4, we also compare graphically the IRFs conditional on a specific shock between sequence-space and state-space solution. All IRFs are extremely close across both techniques also beyond the business cycle horizon.

#### IV. US Business Cycles and Inequality

We apply our estimated model to study the shocks that drive the US business cycle and to understand what these shocks and their propagation imply for the dynamics of inequality. We do so in terms of variance decompositions at business cycle frequency (between 6 and 32 quarters, based on the frequency domain decompositions in Uhlig 2001) and in terms of historical decompositions.<sup>28</sup>

##### A. Variance Decompositions

Figure 1 shows the variance decompositions for output and consumption growth (top panels) for the representative agent model (RANK) as well as for the two heterogeneous agent model estimates (HANK and HANK-X). Further variance decompositions of observables and the credible intervals can be found in online Appendixes A.4 and A.5. Despite the very similar parameter estimates across models, there are some notable differences in the importance of shocks for the business cycle but also many similarities. In all models, technology shocks are the most important drivers of output. This is in line with what other authors have found for the RANK model (Smets and Wouters 2007; Justiniano, Primiceri, and Tambalotti 2010, 2011).

In the HANK model, investment-specific technology shocks are even more important than in RANK because these shocks have a stronger impact on consumption. Positive investment-specific technology shocks move asset prices down and expected wage incomes up; see the impulse response functions in online Appendix A.7. This drives consumption up because wage earners have a larger (intertemporal) marginal propensity to consume than capital holders. More specifically, within our two-asset framework, there is a sizable share of households with little illiquid assets but sufficient liquidity such that they can increase their consumption on impact in the expectations of higher future wages. By contrast, illiquid asset-rich households expect lower asset income in the future, but they smooth this change over a longer period. As a result, consumption goes up on impact. In RANK, this difference of intertemporal MPCs between wage and capital-income earners is absent, and consumption and investment negatively co-move on impact after an investment-specific technology shock. For this reason, investment specific technology shocks add little to consumption movements in RANK.

<sup>27</sup> We calculate the sequence-space solution based on a 300-period transition as in Auclert et al. (2021). We use the IRFs from the state-space solution to obtain the endpoint of that transition in order to eliminate the effect of not being exactly at the stationary equilibrium after 300 periods. We do so because after some of the shocks, the economy has even after 300 periods not fully returned to the stationary equilibrium. This also shows an advantage of the state-space solution over a sequence-space approach, namely, that it can easily deal with highly persistent shocks because there is no need to set a truncation period.

<sup>28</sup> Online Appendix A.7 provides impulse response functions for all shocks and observables.

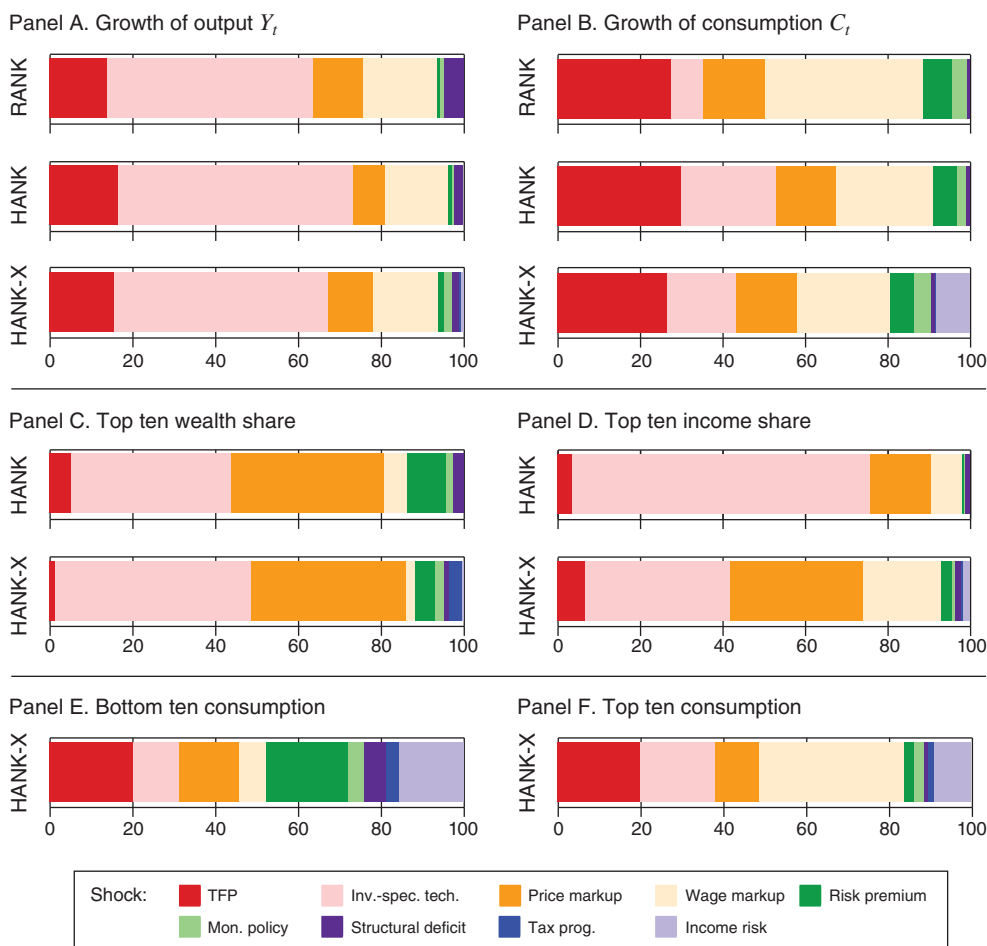


FIGURE 1. VARIANCE DECOMPOSITIONS: OUTPUT GROWTH, CONSUMPTION GROWTH, AND INEQUALITY

*Note:* Conditional variance decompositions at business cycle frequencies (6–32 quarter forecast horizon) for the estimated RANK, HANK, and HANK-X models.

Comparing the variance decompositions of the HANK and HANK-X estimates, we see that the inclusion of inequality data, tax progressivity, and income risk also has some impact on the relative contribution of shocks to aggregate consumption (and to a lesser degree on output). For consumption, demand side shocks become somewhat more important. In particular, we see that uncertainty shocks contribute significantly. However, given the relatively low degree of price stickiness and relatively strong monetary stabilization we estimate, the contribution of all demand shocks to output growth is limited.

The HANK models, different to the RANK setting, make it possible to study the effect of business cycle shocks on inequality; see the middle panels of Figure 1. In fact, the HANK-X model also exploits the inequality time series data for the estimation. Again, we find that technology shocks have an important impact on inequality. However, if we actually use the inequality data for the estimation, the

effect of investment-specific technology shocks in income inequality is tuned down, and markup shocks become more important. For wealth inequality, there is little difference in the effect of technology shocks between the two models. What is more, as the HANK-X variant also allows tax-progressivity shocks to impact on the economy, they gain a visible influence on wealth inequality.

One advantage of the HANK model is that it allows us to think about drivers of consumption throughout the income and wealth distribution. The bottom panels of Figure 1 give an example for this and show the variance decomposition of the average consumption of the 10 percent consumption poorest and consumption richest households. The differences of which shocks are important for these two groups are stark. Consumption of the consumption poor is much more strongly influenced by demand side shocks than consumption of the rich. This suggests that more detailed data on the time series of the consumption distribution might be helpful in identifying business cycle shocks in future work.

### *B. Historical Decompositions of Inequality*

Figure 2 adds to these findings by providing historical decompositions for US inequality dynamics. Recall that we allow for measurement error on inequality when estimating the HANK-X model. This implies that neither the HANK nor the HANK-X variant need to match the inequality data perfectly. The top panels of the figure compare the model-predicted movements of the top ten wealth and income shares with the actual data. Perhaps surprisingly, already the HANK estimate implies a falling wealth inequality during the 1970s, which recovers (and overshoots) in the twenty-first century. The HANK-X model follows the observed wealth inequality closely. The model rather changes the shocks and frictions that drive the business cycle slightly instead of giving up on wealth inequality and “explaining” the data by measurement error.<sup>29</sup> In other words, the cross-sectional data are somewhat informative for the business cycle model, but there is no strong tension.

For income inequality, the situation is different. The HANK models imply a fluctuation of the income share of the top 10 percent that is too volatile. This high volatility is driven by large and procyclical swings in profits. Yet such strong fluctuations are absent in the income inequality data. Surprisingly, however, the model does broadly capture the lower-frequency movements in income inequality with its increase since the 1980s. The figure further suggests that persistent increases in price markup targets are behind those low-frequency movements in income inequality. This finding resembles the evidence by De Loecker and Eeckhout (2020) on the evolution of markups in the United States.<sup>30</sup> Since the cyclical properties of the top ten income share largely follow the cyclical properties of profits in the model, finding too volatile top incomes is intimately linked to the literature that discusses tensions between model-implied markups/profits and the data (see, e.g., Andreasen and Dang 2019; Nekarda and Ramey 2020). The slow-moving nature of top income shares suggests through the lens of our model that profit incomes need to be less

<sup>29</sup>The estimated standard deviations of measurement errors can be found in online Appendix A.3.

<sup>30</sup>There is a growing literature on the rise of markups; see, e.g., Karabarbounis and Neiman (2019); Barkai (2020); Hall (2018); or Kehrig and Vincent (2021).

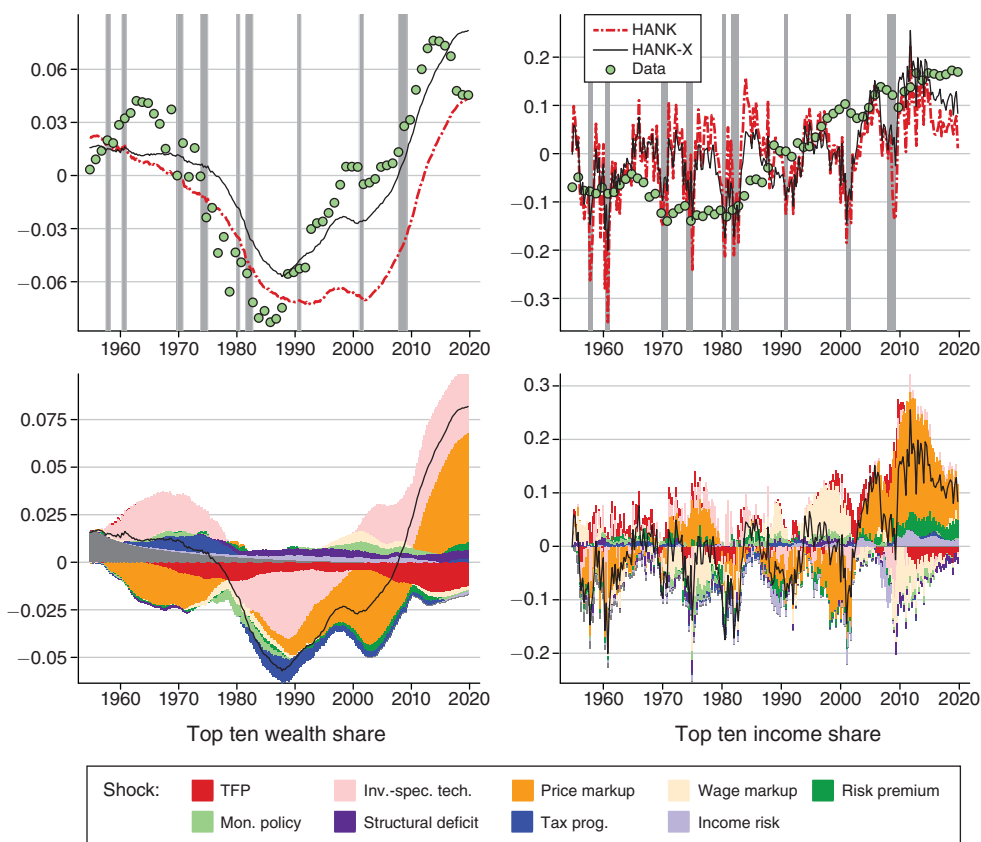


FIGURE 2. HISTORICAL DECOMPOSITIONS: INEQUALITY

*Notes:* Historical decompositions for the top ten wealth and income shares for the estimated HANK and HANK-X models. The top panels show the implied series (HANK and HANK-X) obtained through a Kalman smoother in comparison to the data. The two bottom panels show the actual decomposition into shock contributions for the HANK-X estimates. y-axis: Percent deviation from mean.

volatile or less concentrated among the rich. This holds true despite the fact that we allow the model to choose in the estimation how much surprise changes in profits get distributed to a broader set of households through the market for profit shares. In terms of the HANK-X specific shocks, we see that income uncertainty that drives the dispersion of human capital also has contributed to the increase in income inequality over the last 30 years. This can be viewed as the model's way to capture skill-biased technological change.

When we look at the historical evolution of wealth inequality, we observe that the increase in inequality since its trough in the 1980s is primarily driven by two factors. First, investment-specific technology shocks that have driven up the price of capital. Since the wealthy hold a larger share of their wealth in terms of the illiquid asset in our model, this drives up wealth inequality and resembles the evidence from household balance sheets (Kuhn, Schularick, and Steins 2020). The second-most important factor is again the persistent increase in markups that drives up income inequality and thereby affects wealth inequality, too.

### C. Robustness of Estimation Results

We check the robustness of our estimation results with respect to a number of potentially important modeling choices where other alternatives would have been sensible as well. Concretely, we first estimate the model for the post-Volcker era, implicitly assuming a structural break in 1983. Second, we ask how the HANK estimation changes if we assume a risk aversion of 2 (instead of 4). Third, we model the distribution of union profits such that it leads to no wage compression and the wage markup only affects labor supply but not directly the income distribution. Fourth, we allow (in HANK-X) for a systematic response of tax progressivity to income inequality, and, finally, we assume King, Plosser, and Rebelo (1988) preferences instead of Greenwood, Hercowitz, and Huffman (1988) ones, such that there is a wealth effect in labor supply. Modeling details, estimated parameters, and decompositions are reported in online Appendix B.

We find that the view of the business cycle and inequality dynamics through the lens of the model is relatively robust to these variations, with the exception of the KPR variant. In the post-Volcker period, markup shocks become somewhat more important, in a sense reflecting their larger impact starting in the 1980s that could be seen already in the historical decompositions for the estimation on the full sample. A lower risk aversion leaves the decomposition of output almost unchanged, but investment shocks become less important for consumption and more important for wealth inequality. The effect of wage compression is negligible throughout, with the exception of income inequality; similarly, the estimated systematic response of tax progressivity is weak and thus affects results very little.

While the view of the business cycle is broadly similar between the KPR and GHH variants, marginal data densities clearly prefer the GHH specification both in RANK and HANK.<sup>31</sup> However, the differences become starker when looking at inequality dynamics. Here, the KPR-HANK model does not reproduce the U shape in wealth inequality and misses the trend in income inequality, which makes the rejection of the KPR assumption by the data even stronger than the marginal data densities already indicate.

## V. Conclusion

How much does inequality matter for the business cycle and vice versa? To shed light on this two-way relationship, this paper develops a new reduction technique and provides a toolbox to estimate New Keynesian business cycle models with household heterogeneity and portfolio choice in its state-space representation via Bayesian methods. Concretely, we leverage the fact that we have prior information for model reduction and use this to derive a factor representation of the heterogeneous agent part of the model. The speed and precision of the proposed solution method are comparable to sequence-space approaches suggested by Auclert et al. (2021) and

<sup>31</sup>The (log) marginal data densities for RANK and HANK with KPR preferences are 6,545 and 6,249, respectively, both lower than those of the models with GHH preferences (see Table 2). The drop in the marginal data density when assuming KPR preferences is particularly stark for HANK, where wealth effects on labor supply become even stronger.

Boppart, Krusell, and Mitman (2018). One advantage of the state-space approach lies in the ample tool set that has been developed for state-space models of the business cycle, for example, variance decomposition at the business cycle frequency or historical decompositions.

Using the same set of aggregate shocks and observables as in Smets and Wouters (2007) in the estimation, we find that heterogeneity in household portfolios gives more precedence to technology shocks in explaining consumption at the expense of markup shocks. They increase wages of poor households with high marginal propensities to consume, which leads to more co-movement with investment relative to a setup with a representative agent. When including cross-sectional data and shocks in the estimation, shocks to income risk start to play a role as well—especially for consumption of poor households.

The model successfully replicates the dynamics of US wealth inequality such that there is no strong tension between what business cycle analysis suggests as drivers of the cycle and what the model prefers as drivers of wealth inequality. Today's high inequality is a result of technology shocks driving up asset prices and markup shocks driving up profits, in line with empirical evidence from De Loecker and Eeckhout (2020) and Kuhn, Schularick, and Steins (2020). However, some tension remains between the aggregate time series and income inequality dynamics when it comes to markup shocks. The model predicts that profit incomes and thereby top incomes are too volatile because fitting the aggregates requires large swings in markups in relative terms. This leaves room for future research using inequality data to inform researchers better about the calibration and estimation of the markup process. What is more, we show that different business cycle shocks impact the percentiles of the consumption distribution differently, such that using detailed data on the consumption distribution dynamics might also help to tighten the identification of business cycle shocks in future research.

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